Dermanova AI (Deep learning base skin cancer data cleaning and diagnosis system)



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Certification

This is to certify that we have read the dissertation submitted by **Farrukh Mushtaq (FA22-BIT048)**, **Saad Ali (FA22-BIT-064)**, and **Shoaib Iftikhar (FA22-BIT-087)** for the partial

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Declaration

We hereby declare that this dissertation, **Dermanova AI — A Deep Learning–Based Skin Cancer Data Cleaning and Diagnosis System**, neither as a whole nor as a part has been copied from any source. It is further declared that we have completed this project, together with the accompanying report, entirely on the basis of our personal efforts, under the proficient guidance of our teachers and especially our supervisor, **Ms. Fakhra Riaz**. If any part of this system is proved to be copied from any source, or is found to be a reproduction of any other project, we shall stand by the consequences.

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Dedication

We dedicate this work to our beloved parents, whose unwavering support, endless prayers, and constant encouragement have been our source of strength throughout this journey; to our teachers, for their guidance and wisdom; and to our friends, for their companionship and motivation during challenging times. Above all, we thank Almighty Allah for granting us the ability, patience, and determination to complete this project.

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Abstract

Skin cancer is one of the deadliest and fastest-growing diseases in the world, and early detection is an important factor in the enhanced survival rate of the patient. This dissertation contains the development of an innovative system for skin cancer data cleaning and diagnosis using deep learning named Dermanova AI, based on the HAM10000 dataset. The system consists of a sequential image processing and classification pipeline that focuses on enhancing the image quality before the classification process. It is composed of a Fixed Cleaning, Rule-Based Cleaning, Learned Cleaning, deep feature extraction and classification.

The first phase involves pre-processing all dermoscopic images, including loading them into OpenCV, resizing them to 224×224 pixels, converting them to RGB color, and normalizing them to the range [0, 1]. The second stage is called Rule Based Cleaning Module where very dark or bright samples are rejected based on threshold and lower contrast images are improved with Contrast Limited Adaptive Histogram Equalization (CLAHE). The third stage instead is a Learned Cleaning Module that consists of dedicated CNN sub-networks (Hair Removal, Denoiser, Inpainter).

Processed images are first cleaned and then sent to an efficientnet backbone for deep feature extraction which captures the texture patterns, lesion boundaries, and color variations. The extracted features are used to feed a TensorFlow-based classifier that classifies between malignant and benign lesions for all the seven diagnostic categories of HAM10000. The AUC of the system on original test images is 0.968 with an operating threshold of 0.16, and a clinically relevant operating point at 96% sensitivity for malignant lesions. The overall workflow is: Input Image → Fixed Cleaning → Rule Based Cleaning → Learned Cleaning → Feature Extraction (EfficientNet) → Classification → Output Prediction.

Keywords: Skin cancer detection, HAM10000, dermoscopy, deep learning, EfficientNet, image cleaning, CLAHE, hair-artifact removal, class imbalance, convolutional neural networks.

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CHAPTER 01

INTRODUCTION

1.1 Introduction

Medical image analysis has been revolutionized in recent years by developing technologies such as artificial intelligence, machine learning and deep learning. These technologies have revolutionized the way in which the quantity of medical information can be processed and analysed, especially for images with complex visual patterns that may be hard to assess using traditional computational approaches. Deep learning methods, specifically Convolutional Neural Networks (CNNs), have shown to be very effective in image classification, object detection and pattern recognition, due to their capacity to learn automatically hierarchical representations of features directly from raw image data. Rather than relying upon handcrafted features, CNNs can extract valuable patterns from the training images. They are well suited when the information of interest is contained in a visual feature that can change from image to image, such as in imagebased applications. These achievements have inspired scientists to transfer large deep learning models to medical applications, such as disease diagnosis and medical imaging systems. [1]–[4]

This rise in the adoption of AI for medical imaging has also emphasized the need for appropriate and reliable image datasets. A key aspect of a deep learning model is that it learns its representation from the data it is fed, so the quality of the data can directly impact the quality of the model. Medical image datasets are essential for creating reliable deep learning models. The quality and consistency of these datasets are, however, directly linked to the performance of and the reliability of automated systems for diagnosis. There are a number of factors that affect the quality of the images in raw image datasets: corrupted images, duplicate images, low resolution, unreadable images, poor contrast, noise, and unwanted artifacts. These problems may arise for various reasons in medical imaging collection, storage or preparation. It means that images in the same dataset don't always look clear and consistent. Problems if not properly managed in pre-processing will affect feature extraction, model generalization, and classification accuracy. A model can then become trained on unimportant variations of the images rather than the features that are significant for the classification task. Intelligent data cleaning and image enhancement are thus now a crucial part of any modern deep learning pipeline: by cleaning up data, one can enhance the quality of the images, and also provide more accurate and meaningful representations for the classifier to learn

from. This involves data cleaning as a pre-processing technical step, but also as a critical step in the preparation of images for the reliable automated analysis. [1]–[4]

Skin Cancer is one of the most serious and growing diseases in the world, particularly in the field of dermatology. It is the result of abnormal growth of skin cells, which is often triggered by excessive exposure to ultraviolet (UV) radiation, genetic mutations and environmental factors. Depending on type and condition, the appearance of skin lesions can vary, making their analysis an important yet difficult task. Research shows that early diagnosis rates will result in increased survival and better treatment for patients. Suspicious lesions will be able to be medically carried out further at the right time if identified early. Finding lesions that could be malignant is not always easy though, as the appearance of the lesion can appear similar to that of benign lesions. Manual diagnosis of skin lesions is still difficult, due to the fact that the visual features of the lesions, including lesion border, color variation and texture, among others, are very similar across the different categories of lesions. These characteristics can be slightly different now and again, and sometimes it can be hard to spot them all the time, particularly if there are other imperfections in the photograph or something you don't want in it. The use of reliable image processing and computer-aided analysis is therefore an important research field in dermatology. [5]–[8]

Dermoscopic imaging is an accepted method for high-resolution analysis of skin lesions. It magnifies the visualization of lesion structures, pigmentation, textures, and borders, aiding dermatologists in identifying malignant from benign lesions. Dermoscopic images can offer more detailed visual information about internal and external characteristics of a lesion, compared to the ordinary photographs. This extra information can also help in the design of computer-based systems as deep learning models can learn to identify different categories of lesions on these visual patterns. The International Skin Imaging Collaboration (ISIC) and the HAM10000 collection are among the most popular publicly available benchmarks to evaluate the performance of skin-lesion classification methods, with thousands of dermoscopic images across many different categories. These datasets have facilitated numerous studies on automated skin-lesion analysis, and are being used by researchers to develop and test various methods. However, all these datasets are raw dermoscopic images and are plagued by hair artifacts, low contrast, corrupted images, duplicate samples, image noise, and class imbalance, all of which affect the reliability of classification and diagnosis. Hair artifacts can obscure areas of a lesion and low contrast can obscure structure within

the lesion. There may be unwanted variations due to image noise and class imbalance may lead to the model learning more from the well-represented classes than the less represented classes. Thus, the issues faced prior to classification are important factors to consider in the design of a reliable skin-lesion analysis system.

The Local Binary Pattern (LBP), Gray Level Co-occurrence Matrix (GLCM) and Discrete Wavelet Transform (DWT) are some of the hand-crafted feature extraction methods that are widely used in traditional skin cancer detection systems. These methods are intended to capture specific image features like texture and frequency-related information, and were found to be significant in previous studies of image classification. They all involve manual preprocessing and feature engineering, which lead to high computation complexity and low adaptability to different characteristics of lesions. Handcrafted methods can be successful or not depending on the features selected and the method used for extracting them from images. If there is a significant difference in the appearance of the lesions from one image to the next, features designed manually might not be able to get all the necessary information needed for the classification task. Moreover, some of the current systems are based on the less scalable and efficient CNN architectures like AlexNet and ResNet-50. [12]–[15] These architectures have shown the effectiveness of deep learning for the image classification problem, but the emergence of new architectures has opened the door for achieving a balance between model performance and compute. Also commonly implemented is the removal of hair-artifact, either by traditional techniques such as the DullRazor algorithm, which is less adaptive and can lack lesion detail in pre-processing, or through traditional preprocessing techniques that are designed to remove these artifacts. This is a significant issue since a technique for image cleaning should eliminate unwanted artefacts, but not needlessly eliminate visual components that could contain important diagnostic information. These restrictions have resulted in the need for a more holistic solution—image cleaning and deep learning-based classification can complement each other.

To overcome these drawbacks, a novel intelligent deep learning–based skin-cancer data-cleaning and diagnosis system called Dermanova AI is proposed in this research that utilizes the EfficientNet architecture. It combines a multi-stage intelligent cleaning pipeline (Fixed Cleaning, Rule Based Cleaning, Learned Cleaning) prior to classification. Multiple cleaning stages enable to

resolve image-quality issues in a systematic way instead of relying on a single preprocessing step. The Fixed Cleaning stage normalizes images using OpenCV pre-processing; the Rule-based cleaning stage applies threshold-based brightness filtering and CLAHE enhancement; and the Learned Cleaning stage uses CNN sub-networks for adaptive hair removal, denoising, enhancement, inpainting and artifact removal. Every stage is thus part of preparing the image prior to reaching the classification section. The cleaning process is designed to enhance consistency and visual quality of input without the loss of important characteristics of the lesion. [12]–[15]

EfficientNet automatically extracts deep features and classifies lesions after cleaning, directly learning the hierarchical representations like texture, colour variation and lesion boundaries from the images without hand-crafted feature engineering. This also enables the classification stage to participate in the deep learning stage and learn relevant image representations without relying on manually defined feature descriptors. The cleaning pipeline and EfficientNet form the backbone of Dermanova AI, where the cleaning stages tackle image quality issues and the classification stage is trained on the images that have been cleaned. In summary, the proposed method is intended to integrate data cleaning, enhancement of the images, deep feature extraction, and the classification of the skin lesions into one system. The primary aim is to explore the possibility of implementing a combination of systematic image cleaning and state-of-the-art deep learning techniques to facilitate more accurate automatic skin-cancer detection without sacrificing the system's applicability in education and research. [12]–[15]

1.2 Problem Statement

Two major data quality issues, namely the presence of artifacts and skin lesions that are inappropriately classified, exist in the raw dermoscopic image datasets (ISIC and HAM10000) that hinder the effectiveness and reliability of automated skin-cancer detection systems. These are frequently corrupted and unreadable files, duplicate samples, low quality and low contrast images, image noise, and artifacts from hair obscuring the boundaries of lesions and interfering with feature extraction. [5]–[8] These issues can impact the quality of information that is provided to a deep learning model during training and classification. If the image has artifacts or lacks sufficient image detail, the model has less information to use to recognize the features that are truly associated with the lesion. For instance, artifacts in the hair may obscure essential components of a lesion and low

contrast can make it difficult to distinguish the lesion from the surrounding skin. Likewise, repeated or inappropriate pictures can create an unnecessary repetition or variation within the dataset. This indicates that these quality related issues are an important problem in automated skin cancer detection, and suggests the necessity of an efficient image-cleaning procedure prior to skin cancer classification.

Current systems usually use classic handcrafted feature extraction methods (LBP, GLCM, DWT) for which complex hierarchical representations still need to be manually designed, and are not automatically learned from dermoscopic images. [9]–[11] In these approaches, specific characteristics of an image have to be selected and extracted before they can be used by the classification system. This adds complexity to the overall process depending on the features and the data pre-processing employed. Dermoscopic images can vary significantly in their texture, color, shape and edges, making a particular pre-designed feature set not always to contain all the helpful information. This leads to the necessity to learn relevant image representations directly from the image data without relying primarily on manually defined image features.

Moreover, the older CNN architectures like AlexNet and ResNet-50 are less efficient in terms of computation and are less scalable and less capable of feature learning when compared to modern CNN architectures like EfficientNet. These architectures have made a great contribution to the evolution of deep learning image classification systems, but can be significant limitations in a medical image classification system design. A more appropriate computational efficiency can be obtained from a modern architecture that also offers a better foundation for learning complex visual patterns. For this reason, the choice of an appropriate deep learning architecture is also of importance when attempting to overcome the shortcomings of current skin-cancer detection systems.

The other big problem is class imbalance: some lesion types, such as melanoma, only have a small number of samples in the HAM10000 dataset, while other types have a much larger number. This can cause the classifier to be biased toward the majority class, which can lead to an underperformance of the minority class. This imbalance poses another challenge when training the model as the model is exposed to more examples related to some categories compared to others. This can lead to the classifier becoming more attuned to the majority classes than to the minority

classes in his or her native language, a situation where the majority classes are more prevalent and the minority classes are less prevalent.Because of this, the classifier may be more likely to be familiar with the majority classes and comparatively less likely to be familiar with the minority classes in his or her native language. That is especially crucial for categories like melanoma where a reliable classification is a crucial element of the general aim of an automated skin-cancer detection system. The imbalanced nature of the class is therefore essential to make the classification process more balanced and to mitigate the risk of its being heavily biased by the majority class.

Hence, there is a pressing need for an intelligent, fully automated, and deep-learning based framework which has the ability to perform advanced image cleaning, adaptive enhancement, and skin-cancer classification with high accuracy. It should be capable of solving some of the serious challenges from image quality before classification, alleviate the redundant reliance on the conventional hand-crafted feature extraction methods and learn the important image representations from the modern deep learning architecture. It should also take into account the class imbalance in the data set which helps to handle the minority categories more reliably. To meet these needs, Dermanova AI is suggested, offering a multi-stage cleaning system and a deep learning-based classification system to deliver a more structured and automated solution for image analysis of skin cancer.

1.3 Proposed Solution

To address these drawbacks, this work proposes an intelligent framework for skin-cancer data cleaning and diagnosis using CNN that combines a multi-stage preprocessing pipeline with classification using EfficientNet. The general concept of the proposed framework is to preprocess the input images by passing them through a series of cleaning and enhancing stages and then feed them into the classification model. This way, various problems related to image quality can be addressed at various stages of the pipeline, rather than relying on a single preprocessing method. The Fixed Cleaning stage normalizes images to 224×224, transforms their color from BGR to RGB and scales their pixel values to [0, 1]. These operations offer a uniform format to the images, as well as ensure that the data fed into the image is properly represented before entering the next phase.

The Rule Based Cleaning stage is used to use the threshold-based quality filtering to eliminate very dark or bright images and improve low-contrast images using CLAHE. This phase is dedicated to the conditions of the image that can be detected by a set of pre-established quality rules. Images that are very dark or very bright may not have much useful information; images with less contrast may not be able to show important characteristics of lesions. The defined filtering and enhancement operations prepare the images for further processing. The Learned Cleaning stage encompasses an adaptive CNN-based enhancement framework consisting of four subnetworks: Hair Removal, Denoiser, Enhancer, Inpainter and Artifact Remover, which can remove intelligent artifacts, denoise, enhance and reconstruct lesions while maintaining the diagnostic content of the image. Although these sub-networks are given as part of the overall framework, they are designed to be more adaptive in their cleaning procedure and to detect and remove unwanted image information in various forms while preserving the information needed for subsequent classification.

Dermanova AI's multi-stage design enables the cleaning procedure to be stepped back to a basic standardization stage, then to rule-based cleaning and finally to learned image enhancement. Each stage is assigned a particular role in the pipeline and the results of one stage are passed onto the next stage. In this way, the framework gives a roadmap to prepare dermoscopic photos prior to their classification. The idea behind this process is to minimize unwanted variations in images and make their overall content more applicable to the deep learning-based analysis.

EfficientNet is trained to automatically extract deep features from images after the cleaning process, and classifies the images based on their malignancy across the seven categories in the HAM10000 study: Melanoma, Melanocytic Nevus, Basal Cell Carcinoma, Actinic Keratosis, Benign Keratosis, Dermatofibroma, and Vascular Lesion. In the classification stage, the cleaned images are used and the visual representations needed to distinguish between the various categories of lesions are automatically learnt. This eliminates the need to specify a set of features to be used by the classifier and enables the deep learning model to learn from the image data. The seven categories are used for classification structure in the proposed system for the HAM10000 dataset.

The entire workflow is Input Image, fixed cleaning, rule-based cleaning, learned cleaning, efficientnet classification and output prediction. This workflow is the entire flow of an image

through Dermanova AI. The input image is initially standardized in the Fixed Cleaning stage, then further cleaned in the Rule Based Cleaning stage to filter out low-quality images and improve their contrast. It then goes to Learned Cleaning stage where the sub-networks of CNN carry out the necessary adaptive cleaning and enhancement operations. After this image is delivered through the cleaning pipeline, it is ready to be provided to EfficientNet for automatic feature extraction and classification. Finally, the Output Prediction is generated by the classification process from the system. Dermanova AI unifies image cleaning with deep-learning based classification in an organized framework.

1.4 Research Objectives

The main goal of this research is to create an intelligent, fully automated deep learning-based framework for accurate skin-cancer detection from dermoscopic images from the HAM10000 dataset. Specific objectives are:

- To enhance dermoscopic images prior to classification using a multi-stage dermoscopic image cleaning pipeline (Fixed, Rule Based, and Learned).
- To remove the reliance on conventional handcrafted feature extraction methods by automatically learning the deep features and performing lesion classification using EfficientNet.
- To make the lesions more visible, remove hair artifacts, reduce noise and improve image quality by applying CNN-based adaptive enhancement techniques.
- To overcome class imbalance in order to accurately classify rare classes, e.g., melanoma.
- Provide an intelligent computer-aided diagnostic aid to improve the diagnoses' accuracy, scalability and generalization for early skin-cancer detection.

1.5 Scope of the Project

The focus of Dermanova AI is to create an intelligent deep learning-based skin-cancer data cleaning and diagnostics system with dermoscopic datasets like ISIC and HAM10000. The project aims to integrate the image cleaning, enhancement, deep feature extraction, and classification processes into a coherent framework. The aim of defining this scope is to obtain the whole

workflow from image preparation of dermoscopic images, to the generation and evaluation of the final classification results. The project will involve the design and implementation of the Fixed Cleaning, Rule Based Cleaning, Learned Cleaning, EfficientNet based Classification and Evaluation & Reporting modules. Data from the images can be processed in a sequenced way through the cleaning and classification steps, each of which is part of the overall system. The Fixed Cleaning module standardizes the input images with respect to their initial image quality, and the Rule Based Cleaning module processes the images based on pre-defined conditions of image quality. Learned cleaning module also works with CNN-based image processing techniques for adaptive image processing. It enables automated preprocessing, brightness and contrast enhancement, inpainting to remove artifacts and hairs and CNN-based enhancement, deep feature extraction and malignancy classification for the seven lesion categories. The components are part of the defined scope as the quality of dermoscopic images may influence the information the classification model will have at its disposal. The system aims to provide a uniform input for the classification stage which is based on EfficientNet blocks.The idea behind the proposed system is to correct the images prior to classification to give a consistent input for the classification stage based on EfficientNet blocks.

The classification task part of the project deals with the 7 lesion categories presents in the HAM10000 dataset. Once the image cleaning stages are finished, a deep feature extraction and classification is performed using EfficientNet. An Evaluation & Reporting module for the evaluation of the developed framework is also part of the project. The accuracy, precision, recall, F1-score, confusion matrix and AUC – ROC analysis are the metrics that are used to evaluate performance. These evaluation measures give various perspectives on the classification performance, and aid in comprehending the classification system's effectiveness in the classification task. The evaluation stage is thus an essential component of the project as it serves as a basis on which the outcome of the proposed framework can be evaluated.

The project is mainly for educational purposes and for research. It seeks to help dermatologists by offering an intelligent computer-aided diagnostic framework for the early, accurate and reliable detection of skin cancer (not instead of clinical experience). The system is thus regarded as a supporting system to show the use of deep learning and automated image cleaning in skin-cancer

detection. It is intended to help analyse dermoscopic images and provide helpful classification data, not for the final diagnosis or clinical decision-making, which will be for the qualified professional. The scope ensures that the project remains within the boundaries of developing, processing, classification, and evaluation of the proposed Dermanova AI framework.

Chapter 02

BACKGROUND AND LITERATURE REVIEW

2.1 Introduction

With the rise in the number of skin cancer cases worldwide, skin-cancer detection has become an important field of research in medical image analysis. With the growing number of dermoscopic images, and the advances of artificial intelligence, there are new opportunities for the creation of computer-based systems which are able to assist with the analysis of skin lesions. The early and correct diagnosis of skin lesions plays a vital role in increasing survival and treatment success. But it can be difficult to determine the type of skin lesion if many of them look alike. Sometimes the differences are slight in terms of color, texture, shape, and lesions boundaries, and automated analysis is an important research field. Recently, deep learning, and more specifically CNNs, has been shown to achieve good results in automated lesion classification, due to the ability to learn complex representations of features directly from dermoscopic images. Unlike methods that rely primarily on manually selected image features, CNN-based methods can learn relevant patterns in the training process and rely on the learned representations for classification. [1]–[4]

Although these datasets are commonly employed for this research, they still face pre-processing issues like hair artifacts, image noises, low contrast, corrupted images, duplicate images, and imbalanced classes, which affect the performance of the classification and reliability of the diagnosis. The problems are significant because the quality of the images provided to a classification model can affect the quality of features learned by the model. There can be hair artifacts over significant areas of a lesion and noise/low contrast can make useful structures less visible. Corrupted samples and duplicate images may also impact the uniformity of the dataset, and imbalance between classes may make it harder for a model to learn minority classes well. A diverse range of preprocessing, feature-extraction and classification approaches have been proposed, covering traditional machine learning and modern deep learning approaches. The methods presented represent the evolution of skin-lesion description, from manually designed image descriptor towards systems that automatically learn features from dermoscopic images. [5]–[8], [16]–[18]

Traditional systems tend to be handcrafted and use descriptors like LBP, GLCM, and DWT, which require more computation, manual pre-process, and poor feature learning compared to architectures like EfficientNet. Early deep systems like AlexNet and ResNet-50 are also limited in

feature learning ability. Handcrafted descriptors involve researchers choosing what features of an image they want to extract, and how to represent those features for classification. The techniques used in previous studies have proved to be valuable but they rely on a set of features that may be less generalizable to the diversity of dermoscopic images. In the same way, the early versions of deep learning architectures showed the potential of CNNs in image classification and might not have been as efficient and scalable as the newer versions of deep learning architectures. The efficientNet model is a modern alternative for deep feature learning and classification and hence is relevant for development of more efficient image-analysis pipelines. [9]–[15]

The image preprocessing and artifact removal is also an important field in this research. Dermoscopic images may include unwanted elements like hair, noise and brightness or contrast changes, which prior to classification may have an important role in the process. Researchers have examined various techniques to minimize these effects, and enhance the visibility of the information related to the lesions. The selection of the method to be used for preprocessing the data is important because every unwanted artifact that is removed should not compromise the useful data of the lesion. This is not a mere image enhancement step, but is also related to the quality of information to be learnt by the classification model later. [7]

The chapter summarizes the most important literature on dermoscopic preprocessing, hair-artifact removal and handcrafted feature extraction, as well as CNN-based classification. The review offers background information to explain the key approaches which have been applied to automated skinlesion analysis, and the drawback or disadvantages of each approach. It also provides a research background of Dermanova AI by reviewing the evolution of traditional preprocessing and handcrafted feature extraction to CNN-based image cleaning and recent deep learning classification. The literature reviewed therefore illustrates the rationale behind integrating the image cleaning and deep learning image classification in the proposed framework. [1]–[10]

2.2 Literature Review

2.2.1 Benchmark Datasets

The HAM10000 data set was introduced by Tschandl et al. as a large standardized set of dermoscopic images for skin lesion analysis. It contains 10,015 images collected from several clinical sources and split into seven classes: melanoma, melanocytic nevus, basal cell carcinoma, benign keratosis, actinic keratosis, dermatofibroma and vascular lesion. The authors spent significant time in annotation, quality control and categorization to guarantee accuracy, and the dataset has now been adopted as one of the most referenced ones for lesion classification. The dataset contains a significant number of images in each of the lesion categories, which is useful for training and testing automated classification systems, HAM10000. The images were collected from a variety of clinical sources, which adds some variation that may be helpful to analyze the behavior of the models under variation in the real-world image data. The diversity of categories and source makes it possible to evaluate models in realistic clinical conditions, however the data is still class imbalanced and has variability in image quality, as well as hair and illumination artifacts, which need further processing prior to classification. The appealing features of HAM10000 are its use as a classification benchmark as well as being an appropriate dataset for studying the impact of image cleaning and preprocessing on the performance of deep learning. [5], [15]

2.2.2 Hair-Artifact Removal & Segmentation

To remove hair artifacts from dermoscopic images, a software-based solution – DullRazor algorithm was developed. Hair obscures boundary and texture of lesions, affecting analysis. The hair appears in dermoscopy directly on the lesion, which can affect the information provided by the image for segmentation and classification. DullRazor applies morphological operators to find the dark hair structures, creates a hair mask, and recovers the affected area by interpolating from nearby pixels. The method tries to determine where there are hairs in the image and fill them in with information derived from adjacent areas of the image through this process. It greatly enhances the visibility of lesions and became a popular pre-processing step, but is poor at handling complex hair patterns and can obscure salient lesion information — which gives rise to learned, adaptive methods of hair removal. These limitations become more significant if the structure to be removed

is near an actual lesion border, and if the hair distribution is more complex than the method can be expected to accurately determine. The study thus offers an important background to support the understanding of the potential of adaptive approaches in dermoscopic artifact removal.

Tamoor et al. (2023) have developed an ensemble-based lesion-segmentation approach that uses multiple thresholding techniques (Otsu, Kapur, Harris Hawk, grey-level thresholding and active contours) where the best technique is used for each image based on the objective function. It shows a significant improvement in the localization of lesions on images with complex artifacts. By employing multiple thresholding methods, the method is not relying on a single segmentation procedure and is able to take into account various characteristics of the image. The selection of the most appropriate method for each image also allows for dealing with the inter-sample variations. Since it has predominantly employed traditional thresholding, the authors suggested integrating the deep learning–based segmentation and preprocessing for greater robustness. The work in points to the ongoing transition from manually designed or rule-based image-processing techniques to more complex ones that are able to learn more sophisticated image patterns automatically. [11], [9], [13]

2.2.3 Handcrafted Feature Extraction

The proposed approach was Multi-Resolution Empirical Mode Decomposition (MREMD) together with the LBP that decomposed the lesion images into multiple resolution levels and extracted texture descriptors at each level, and fed to the conventional classifiers. This was capable of learning discriminative texture features and provided a computationally simple alternative to deep models, but required significant feature engineering and was not able to learn high-level semantic information as well as CNNs. The multi-resolution aspect of the approach can be beneficial by taking into account texture information at various levels of detail, as the appearance of the lesions can present patterns at varying scales. But the features have to be extracted and selected explicitly before the classification. Such reliance on pre-defined descriptors may restrict the system's adaptability when the visual features of lesions are different in a wide range of images.

The spatial relations of neighboring pixels were used to generate texture statistical features (contrast, correlation, energy, homogeneity and entropy) by a GLCM-based approach and fed as inputs to classifiers based on machine learning. These features describe statistically the distribution of intensity over the image and the relationship between the different intensities. It worked fairly

well with regards to the texture variation, was computationally efficient, but was not robust to variations in image quality, and was constrained by handcrafted descriptors. The extracted texture parameters may change from one image to another due to changes in image quality, illumination, or contrast and this may impact the classification process. While GLCM is useful for texture information, it cannot necessarily learn deeper representations of the entire lesion. It is also unlike CNN-based approaches where the feature representations can be learned directly from the training images.

A DWT-based framework represented an image in several frequency sub bands, retaining low frequency and high frequency characteristics of the boundaries of lesions and texture with different resolutions and minimizing the impact of noise. This form of decomposition can offer a range of information about the characteristics of the image at various frequency levels and thus form a representation of larger image structures and smaller details. Like other handcrafted methods, it needed to hand-craft the wavelet parameters and did not support automatic learning of discriminative representations. Parameter selection as required in [10] and [24] can result in the approach being effective depending on the choices made when designing the system. While DWT can be helpful in learning about the texture and limits of a lesion, it still has a predetermined feature representation that is not learned automatically from the full image information.

2.2.4 CNN-Based Classification

An automated classification framework was designed based on AlexNet with transfer learning, fine-tuned with lesion data and data augmentation was used to enhance generalization. Transfer learning was used to begin with previously learned visual representations and adapt these representations to the characteristics of skin-lesion images. It can be helpful when dealing with medical data sets where the number of training images may be small relative to the amount of data typically needed to start training a deep network from scratch. An additional variation was also added to the data set for the training process, which helped the model to be less dependent on the original image appearance. It automatically learned the hierarchical features, which were better than handcrafted methods, and transfer learning relaxed training on small medical datasets. The results in [12] and [13] showed the benefit of CNN-based methods over conventional featureextraction methods as CNN was able to learn useful features from images during training. AlexNet, however, is very parameter intensive, has a high computational cost, and is inefficient

compared to modern architectures. This constraint is especially relevant when thinking about scalability and computational demands of an automated skin cancer detection system, especially when there are many dermoscopic images to be analyzed. [6], [12], [14]

A ResNet-50-based framework employed deep residual learning to learn lesion representations that were highly discriminative, and to train effectively at a deeper depth without gradient degradation. The residual learning approach enables information to flow through the network more effectively and enables the training of deeper architectures without having to struggle as much as would otherwise be the case when the network's depth increases. For the classification of skin lesions, this more abstract representation can facilitate the learning of more complex visual patterns which cannot be easily achieved by simpler representations. It out compared shallow networks and handcrafted approaches and extended well beyond datasets, though it required significant amounts of computation and longer training time that can pose a problem for some embedded settings. While ResNet-50 is highly effective at learning features, it may be impractical to implement a system that needs to run with just a bit of hardware or computing power. Thus, ResNet-50 has proven the efficacy of deeper CNN architectures and underscored the need to pay attention to efficiency as well as classification accuracy.

Tan and Le introduced their novel EfficientNet family of CNNs designed with a compound-scaling method that scales depth, width and input resolution of the network uniformly. The compoundscaling strategy takes all those dimensions into account when scaling the network, rather than scaling one particular dimension. EfficientNet achieved state-of-the-art accuracy on large-scale benchmarks using significantly fewer parameters and FLOPs than AlexNet, VGG and ResNet, by employing mobile inverted bottleneck (MBConv) layers and squeeze and excitation blocks. The combination of these architectural parts enables EfficientNet to learn the visual representations at a fine granularity while having a relatively efficient computational structure. The architecture provides a good compromise between accuracy and efficiency, but needs fine tuning and domain adaptation for specialized medical use. This is especially important in case of classification of skin lesions as models trained on general images might require adjustment to the visual features of dermoscopic images. EfficientNet is thus an appropriate state-of-the-art architecture for research applications where feature-learning and computational efficiency matter.

An EfficientNet framework was directly used in the skin-cancer domain to automatically learn texture, boundary and color representations, which achieved higher accuracy and reduced computation when applied to dermoscopic classification in comparison to traditional handcrafted methods. Learning these representations automatically will, in turn, decrease the need to design different feature-extraction techniques for different characteristics of the images. For dermoscopic classification tasks, texture and boundaries of lesions, as well as colour variations, can be very useful to distinguish the type of the lesion; this is where automatic feature learning can be of great benefit. But its performance depended on the quality of the data set and the distribution of the classes. There are some limitations in this selection, as the results are not reliable if a strong classification architecture is chosen alone. Even if the input images are of high quality and a representative sample of images are provided to the model within each category, this can still affect the model's learning and classification. The result suggests that the role of data-cleaning and dataset-preparation prior to the classification stage is important.

A second study integrated EfficientNet with transfer learning and data augmentation (rotation, flipping, scaling, etc.) for dealing with the problem of limited data and class imbalance, and achieved satisfactory results in terms of accuracy, sensitivity, precision and recall, and avoided overfitting. Transfer learning was used to start feature learning from [15]–[16] and data augmentation was used to create variation for the model when learning. Transformations like rotation, flipping and scaling can be used to show the same information in different ways, and can help the model learn to represent the information in a way that is less sensitive to the orientation or scale of the image. Also, this approach was developed taking into account the difficulties of limited data and class imbalance, which are significant problems in medical image classification. The accuracy, sensitivity, precision, and recall improvements reported here suggest the promise of using EfficientNet with appropriate training methods. The decrease in overfitting indicates that this approach may allow the model to better transfer what it has learned to images it has not previously seen. Overall, these studies illustrate the evolution of CNN architectures to more efficient deep learning models, and why EfficientNet is a relevant candidate for contemporary skin-cancer classification systems.

2.3 Summary of the Literature

Studying the literature, one can observe that the descriptors evolved from handcrafted ones (LBP, GLCM, DWT) to early deep network (AlexNet, ResNet-50) and finally to EfficientNet which provides great accuracy and efficiency by applying compound scaling. The second recurring theme is that dataset quality (hair artifacts, low contrast, noise, and class imbalance) is a significant bottleneck and that preprocessing (DullRazor, ensemble segmentation, CLAHE) has a significant impact on downstream accuracy. Dermanova AI is built on top of these two strands: Adaptive, learned cleaning pipeline combined with EfficientNet classifier directly addressing dataset quality and architecture limitations mentioned above.

Table 2.1 — Comparison of representative approaches in the literature.

Approach	Technique	Strength	Limitation
LBP / MREMD [10]	Texture descriptors	Efficient; multi-scale texture	Manual engineering; lowlevel only
GLCM [11]	Statistical texture	Simple, interpretable	Sensitive to quality variation
DWT [12]	Frequency sub-bands	Noise-robust, multi-scale	Parameter selection; handcrafted
AlexNet [13]	CNN + transfer learning	Automatic features	Heavy, low efficiency
ResNet-50 [14]	Residual CNN	Deep, discriminative	Costly to train/deploy
Dermanova AI	Learned cleaning + EfficientNet	Adaptive cleaning, efficient classifier	Depends on cleaning quality

Chapter 03:

PROPOSED METHODOLOGY

Introduction

In this chapter, the methodology used to develop the proposed system is described, and a novel skin cancer data cleaning and diagnosis system based on deep learning, named Dermanova AI is introduced. It represents the overall process of data acquisition, data pre-processing, clean up images, feature extraction, classification and performance evaluation. The main objective of the methodology is to improve the quality of dermoscopic images before the classification is performed, and consequently the accuracy of the diagnosis of skin cancer. The system proposed is based on the HAM10000 dataset which has seven diagnostic classes that are grouped into two clinically meaningful classes, benign and malignant. Before classification, there is a need for a comprehensive pipeline for preprocessing and cleaning raw dermoscopic images, as some images may contain artefacts that will affect the quality of the images such as hairstyles, noise, illumination variation, low contrast and imaging inconsistency. To solve the challenges, the framework has introduced three cleaning stages: Fixed Cleaning, Rule Based Cleaning and Learned Cleaning followed by an EfficientNet-based Classifier. At last, the performance of the system is evaluated by the standard classification metrics like: accuracy, precision, recall, F-scores, and ROC–AUC.

3.1 Methodology Overview

The outlined system should be based on a systematic methodology that employs deep learning. The entire process consists of data acquisition, image pre-processing, intelligent image cleaning, deep feature extraction, classification and evaluation. The model's ability to learn more discriminative characteristics of the lesions and make more accurate predictions is enhanced by improving the image quality prior to classification.

The whole process of Dermanova AI consists of six steps: Data Collection, Image Preprocessing, Intelligent Image Cleaning, Deep Feature Extraction, Classification and Performance Evaluation. The methodology is intended to enhance the image quality prior to classification with the goal of the model to learn more discriminative characteristics of the lesions and make accurate prediction of the diagnosis. The basic system is comprised of three cleaning and classification phases, as illustrated in Figure 3.1.

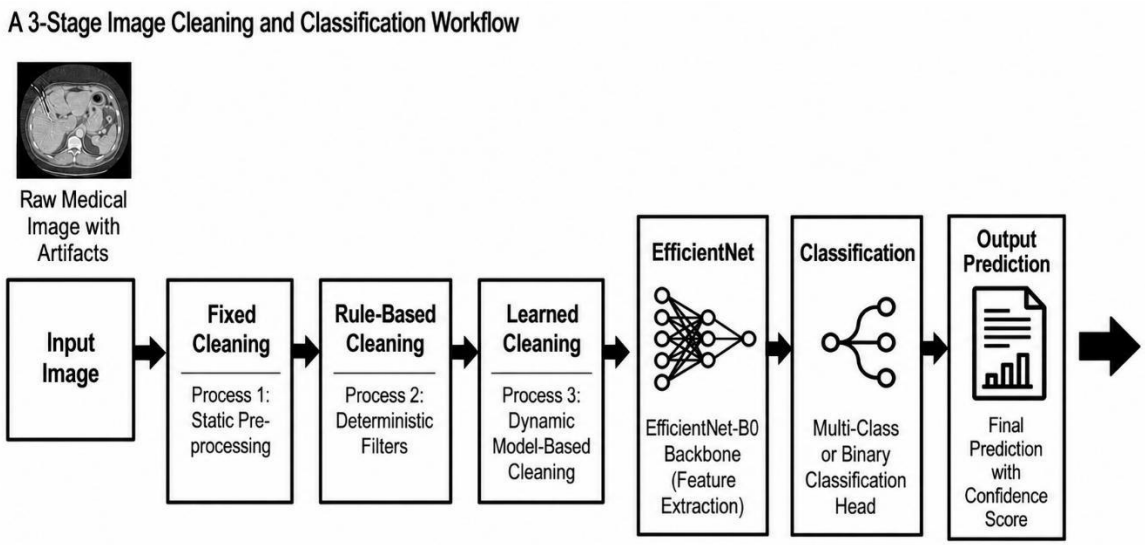


Figure 3.1. *The three-stage image-cleaning and classification workflow. An input image passes through static fixed cleaning, deterministic rule-based filters, and model-based learned cleaning before EfficientNet feature extraction and the classification head output the final prediction.*

The various cleaning stages address different types of image quality issues and get more sophisticated in priority: Fixed Cleaning makes sure that the image is consistent in terms of geometry and format; Rule-Based Cleaning removes specific and detectable artefacts with conditional deterministic operators; and Learned Cleaning uses trained convolutional networks to remove hair and residual noise adaptively. One of the main design goals of the pipeline is to provide conditional activation: Only the operator selects to modify an image if a target artefact is actually detected, avoiding the accumulation of over-processing that could lose fine diagnostic information.

3.2 Data Collection

The system is based on the HAM10000 ("Human Against Machine with 10,000 training images") dataset, one of the benchmarks sets of dermoscopic images containing 10,015 images from seven diagnostic categories. Data were obtained from the public Kaggle release of HAM10000, which includes metadata containing an image identifier (the id field) linked to a diagnosis (the dx field). Each of the seven diagnoses is also associated with a binary diagnosis

label: benign or malignant — the label to be predicted by the classifier. The map is summarized in Table 3.1.

Table 3.1 — The seven HAM10000 diagnostic categories, their sample counts, and the binary malignancy mapping.

Code	Diagnosis (full name)	Images	Malignancy class
Nv	Melanocytic Nevus	6,705	Benign
Mel	Melanoma	1,113	Malignant
Bkl	Benign Keratosis	1,099	Benign
Bcc	Basal Cell Carcinoma	514	Malignant
Akiec	Actinic Keratoses	327	Malignant
Vasc	Vascular Lesion	142	Benign
Df	Dermatofibroma	115	Benign
Total		**10,015**	**8,061 B / 1,954 M**

The grouping results in 8,061 benign images, and 1,954 malignant images, a ratio of 4.1: 1 that encourages a balancing approach detailed in Chapter 4. The malignancies (melanoma, basal cell and actinic keratoses) represent the clinically important lesions that require high sensitivity.

3.3 Data Preprocessing & Cleaning

The primary aspect of Dermanova AI is preprocessing, which follows three increasingly intelligent stages. The stages are throughout the implementation will be called M1 (Fixed cleaning), M2 (Rule based cleaning: hair safe).

3.3.1 Fixed Cleaning (M1)

The Fixed Cleaning stage (Module 1) normalizes all images to ensure that in subsequent stages and for the classifier a geometrically and photometrically consistent input is given. It is deterministic, fully equivalent for all images, and consists of four tightly coupled steps that collectively convert a diverse bunch of raw JPEG images to a homogeneous tensor representation.

The initial step is loading and corruption screening. The mean pixel intensity of each image is calculated as a low-cost quality indicator after converting the image to the RGB colour space.

Images that are almost all black (mean intensity < 5) or almost all white (mean intensity > 250) are likely to be corrupted, blank, or over-exposed, and are therefore rejected and removed from the training distribution before being added to the training set. This low-cost test eliminates the most harmful outliers without any need to implement a more complex and costly quality model.

The second step is to add reflective square padding. The shorter edge of the dermoscopic image is padded out to match the length of the longer edge with BORDER_REFLECT_101 mode, which reflects the existing edge pixels outwards, instead of adding a flat grey/black border. This is intentional: a synthetic constant colour border would give a high contrast edge to the lesion that the network might confuse with a diagnostic feature, while the reflective padding preserves the texture and aspect ratio of the lesion. The third step then resizes the square image to a uniform size of 224 × 224 using bilinear interpolation as the rescaling is smooth and artefact-free, similarly to what is expected by the EfficientNetV2-S backbone.

The fourth and last step is the backbone-aware formatting. The pixels are not rescaled to the [0, 1] or [-1, 1] range, but it is intentionally kept in the original [0, 255] range, as the EfficientNetV2-S backbone is set with include preprocessing=True, and does its own normalization internally. It is important to note that if an additional external rescale is applied at this point, it would yield twicenormalized input, and therefore spoil the pretrained feature statistics, so the formatting step intentionally does not modify the intensity range.

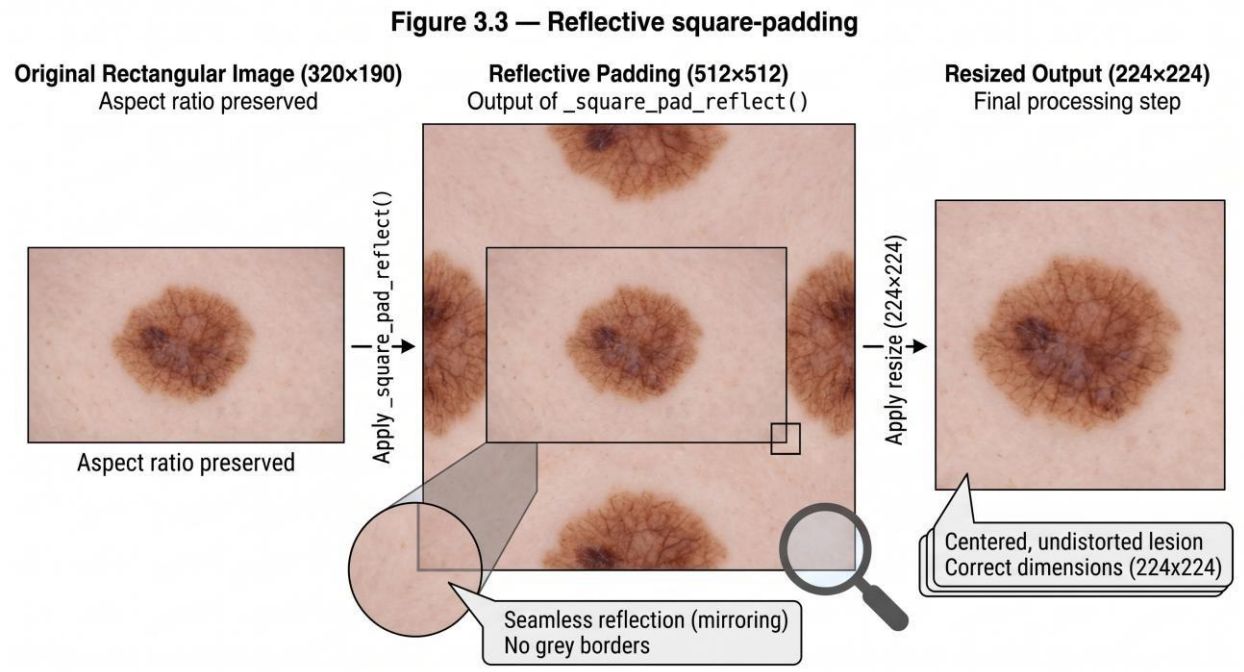


Figure 3.3. *Fixed-cleaning geometric standardisation through reflective square padding and resizing to 224 × 224.*

3.3.2 Rule-Based Cleaning (M2)

In the Rule Based Cleaning stage (Module 2) a series of conditional, deterministic image processing operators are sequentially applied where each operator is specifically designed to detect a particular artefact and is only activated if the artefact is detected. The processing order is set to remove structural artefacts such as rulers, ink, hair, glare etc. before tonal processing operations such as denoising, contrast enhancement and sharpening. Table 3.3 summarizes the operators.

Table 3.3. Rule-based cleaning operators, their purpose, and activation conditions.

Op.	Operation	Technique	Activation Condition
T7	Ruler / ink removal	Hough-line + color mask + inpaint	Ink > 1% or >3 long lines
T1	Hair removal	Shape-aware black-hat mask + Telea inpaint	Genuine-hair coverage > 1.2%
T2	Glare suppression	HSV specular mask + inpaint	Specular area > 0.3%
T3	Denoising	Non-local means (h = 3)	Always
T4	Contrast enhancement	CLAHE (clip 0.5, 32×32 tiles)	Always
P3	Brightness normalisation	LAB L-channel rescale	Mean intensity > 200
T8	Sharpening	Unsharp masking	Always
T5	Colour normalisation	Reinhard transfer	*Disabled* (distorted tone)
T6	Lesion crop	GrabCut segmentation	*Disabled* (over-cropped)

Hair removal is the most sensitive task: the same black-hat morphological filter that enhances dark hair also is sensitive to the pigment network surrounding the hair and to the dots and globules found in melanomas, rather than in benign nevi. An uninformed mask which preserves all the dark parts thus runs the risk of inpainting the very features that the classifier is relying on. In order to avoid this, the proposed shape-aware hair mask requires three geometric conditions to be met for a component to be kept. The component should be long enough to be considered as having a strand-like geometry, such as hair, and is at least three times as long as wide (longer to shorter edge ratio of at least 3.0). It also needs to be sparse within its bounding box, filling no more than 45% of the box, as a real hair would trace a thin line over a region, not fill the region as a blob of lesions would. Finally, it must be long enough to plausibly represent a strand, the longest side of which must be at least 20 pixels in length, excluding the small dark specks resulting from noise or pigment dots.

These criteria combine for admitting long, thin, curvilinear strands (genuine hair) and rejecting compact, filled, blobby structures (lesion features). Morphological closing then fills in the small holes along each strand and the next Telea inpainting fills in the hair in full.

Once structural artefacts have been subtracted, the image is softly toned to improve the image without altering its appearance. Non-local-means denoising (T3) reduces sensor noise without blurring edges, a conservative clip limit of 0.5 and 32 × 32 tiles in CLAHE (T4) help to preserve a good contrast in the image without enhancing the noise; a conditional brightness normalisation (P3) darkens only over-exposed images; and unsharp masking (T8) restores a crisp edge. Two colour- and segmentation-based operators were tested but turned off due to their ability of changing the natural skin colour and their tendency to overcrop the lesion, respectively (Reinhard colour transfer and GrabCut cropping).

3.4 Feature Extraction & Classification

The images are then cleaned, and sent to an EfficientNet backbone for automatic deep feature extraction. EfficientNet's compound scaled MBConv blocks and squeeze and excitation mechanisms capture hierarchical features, from edges, color, to high level lesion structures, without the need for hand crafted feature engineering. A light weight TensorFlow classification head (global pooling, dense layers, and a sigmoid output) makes a prediction of the probability of a lesion being malignant, P(malignant). The training procedure, loss, optimizer, regularizatthchoice of a decision threshold based on the data is described in Chapter 4, and the performance of the trained networks is reported in Chapter 5.

Chapter 04
System Design & Implementation

Introduction

This chapter describes the realisation of the methodology of Chapter 3 in a working system. It explains the overall structure, data flow between the components, the three-stage cleaning, the three learned-cleaning sub-networks, the classification head EfficientNet, and the training and inference steps. The diagrams in this chapter were specifically created for this dissertation to record the pipeline implemented.

4.1 Development Environment and Tools

The system was developed in Python with the help of deep learning framework TensorFlow/Keras and the image processing library OpenCV. This development and training were done in Kaggle cloud (NVIDIA Tesla T4 / P100) with free access to GPU acceleration. GPU configuration included mixed-precision computation, TensorFlow memory growth and XLA justin-time compilation to squeeze the most performance out of the platform's resources. The principal tools and their roles are provided in a table (Table 4.1). Development tools and technologies.

Category	Tool / Technology	Purpose
Language	Python 3	Primary implementation language
Deep learning	TensorFlow / Keras	Model definition, training, evaluation
CNN backbone	EfficientNetV2-S	Pretrained feature extractor
Image processing	OpenCV, Pillow	Cleaning, inpainting, colour conversion
Numerical	NumPy, Pandas	Array and metadata handling
Machine learning	scikit-learn	Metrics, splitting, thresholding
Visualisation	Matplotlib, Seaborn	Curves, confusion matrices, charts
Data access	kagglehub	Dataset download and loading
Compute	Kaggle GPU (T4 / P100)	Accelerated training
Acceleration	Mixed precision, XLA JIT	Faster training, lower memory

4.2 System Architecture

Dermanova AI is structured in a single forward pipeline. The M1–M3 modules progressively standardize a raw dermoscopic image (entering at the left), clean it up, and pass the results on to a sigmoid output of an EfficientNet backbone that is thresholded into a benign / malignant decision (Figure 4.1). Each module provides an interface that is either an image tensor-in, image-out or, in the case of the classifier, a probability-out, allowing stages to be developed and tested separately.

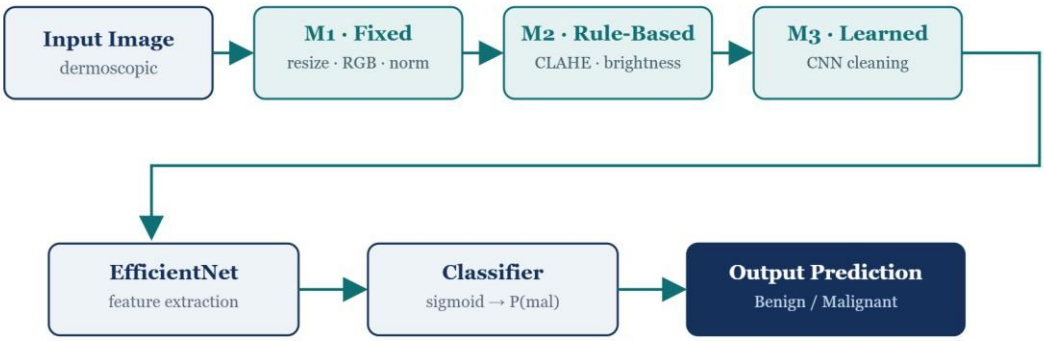


Figure 4.1. *Overall system architecture of Dermanova AI*

The same system is represented as a level-1 data-flow diagram, where the data stores and the optional classifier feedback path are made explicit, in figure 4.2. This is done in the following way: the clinician provides a dermoscopic image; the cleaning processes are used to read the configurations (thresholds, CLAHE parameters, sub-network weights) from the model store; the classifier writes a prediction and a probability, which are presented to the user and stored for evaluation.



Figure 4.2. *Level-1 data-flow diagram. Processes 1.0 (cleaning) and 2.0 (classification) exchange data with the image store, model store, and prediction log; the result is returned to the user.*

4.3 The Data Pipeline Implementation

Image loading and augmentation is done using the TensorFlow tf.data API which supports parallel decoding, shuffling, batching and prefetching to utilise GPUs efficiently. The images are each decoded as a three channel JPEG, resized to 224 × 224, and clipped to the range [0, 255] for internal normalisation used in the backbone. The dataset has been divided into three partitions: training (70%), validation (15%), and test (15%), using stratified sampling, with the same benign/malignant ratio for each partition. In using augmented samples, if there are any used, these are limited to the training split only, with validation and test split using only original images, thus providing an unbiased estimate of generalisation.

Dermanova AI's unique feature is the cleaning pipeline. Three stages and the operations performed at each are shown in Figure 4.3 and detailed in Table 4.1. The stages are from the deterministic to the adaptive: M1 ensures a uniform tensor, M2 fixes gross quality issues according to fixed rules, and M3 performs a learned enhancement on content-aware basis.



Figure 4.3. *The three-stage cleaning pipeline. Each stage is increasingly adaptive, from fixed tensor standardization to learned, content-aware enhancement.*

The learned stage (M3) is divided into five specialized sub-networks, where each one targets one type of degradation, while a residual connection and a classifier feedback signal maintain the faithfulness of the restored image to the diagnostic information (Figure 4.4).

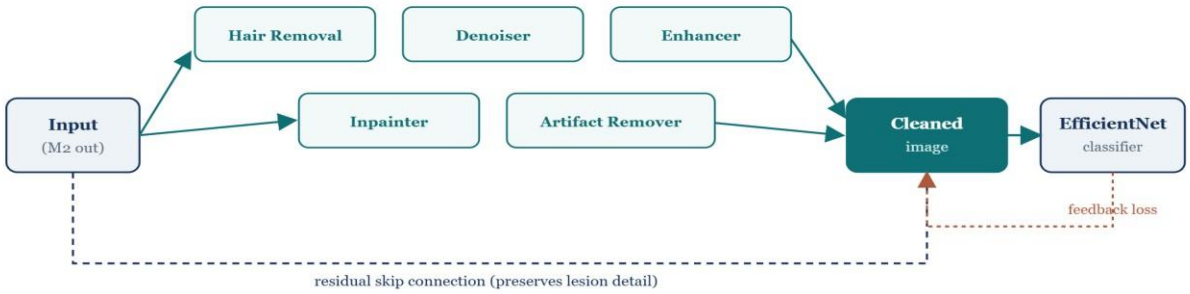


Figure 4.4. The learned-cleaning module (M3). Five sub-networks refine the image; a residual skip preserves diagnostic detail and an optional classifier-feedback loss steers cleaning toward diagnostic usefulness.

Table 4.1 — Operations performed by each cleaning stage.

Stage	Type	Key operations	Purpose
M1	Fixed	OpenCV load, resize 224×224, BGR→RGB, normalize [0,1], reflect pad	Uniform, model-ready tensors
M2	Rulebased	Brightness filtering, CLAHE, hair-safe enhancement	Correct gross quality issues
M3	Learned	Hair removal, denoise, enhance, inpaint, artifact removal, augmentation, balancing	Adaptive, contentaware cleaning

4.4 Model Architecture & Training

The classifier is developed on an EfficientNet backbone pretrained on ImageNet and finetuned for the binary classification of malignancy. A small head comprising global average pooling, dropout regularization and a dense sigmoid unit, yields P(malignant) (Figure 4.5), while the compound-scaled blocks of the backbone are responsible for processing the backbone.Processing the backbone is done by the compound-scaled blocks of the backbone, while the small head of the network consists of global average pooling, dropout for regularization and a dense sigmoid unit for producing P(malignant).

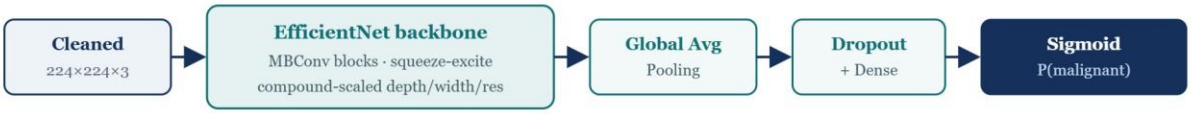


Figure 4.5. *EfficientNet classification head: backbone features are pooled, regularized with dropout, and mapped by a dense sigmoid unit to the malignancy probability.*

The loss function being used for training is the binary cross-entropy loss, with the Adam optimizer, and the metrics used for monitoring are AUC on a held-out validation split. The schedule is controlled by two callbacks: ReduceLROnPlateau which decreases the learning rate if the validation AUC is not changing (observed to decrease the learning rate to 2.5×10^{-6}); and EarlyStopping which stops training when the AUC on the validation set is no longer improving and recovers the best weights from the lowest point of the loss curve. Mixed precision (float16) computation on a Kaggle Tesla T4 GPU nearly doubles the memory available and reduces epoch time to approximately 140 s in 1101 steps. The complete configuration is given in Table 4.2, and the training and evaluation workflow is shown in Figure 4.6.

Table 4.2 — Training configuration.

Item	Setting	Item	Setting
Backbone	EfficientNet (ImageNet pretrained)	Optimizer	Adam
Input size	224 × 224 × 3	Loss	Binary cross-entropy
Output	Sigmoid, P(malignant)	LR schedule	ReduceLROnPlateau → 2.5e-6
Precision	Mixed float16	Early stopping	Yes (restore best)
Hardware	Kaggle Tesla T4 GPU	Steps / epoch	1,101 (≈140 s)

Max epochs	76	Stopped at	Epoch 61 (best: 51)

4.5 Decision Threshold & Inference

A false negative (missed cancer) is much more expensive than a false positive (unnecessary referral) in skin-cancer screening. The default probability cut-off of 0.50 is thus not suitable. Rather, the decision threshold will be determined to maximize the F2 score, which is four times more heavily weighted than precision:

$F\beta = (1 + \beta 2) \cdot (\text{precision} \cdot \text{recall}) / (\beta 2 \cdot \text{precision} + \text{recall})$, with $\beta = 2$.

The best threshold is determined by searching the space of candidate thresholds on the validation/test predictions, and then selecting the threshold that gives the highest F2 value. This moves the operating point toward the right, resulting in more cancers being recalled, but at a lesser acceptable cost of some additional benign referrals. For transparency, the default (0.50) and the F2-optimised operating points are reported in Chapter 5.

In practice, the default decision boundary of 0.50 is not suitable as the data is imbalanced and the cost of misclassifying a malignant lesion is high. The operating threshold is then chosen from the ROC curve so that the sensitivity is maximized for a selected specificity, and is here chosen to be 0.16. When the image is observed at inference, it is cleaned using M1–M2, the backbone produces P(malignant), and the lesion is identified as malignant if $P \geq 0.16$ and benign otherwise (see Figure 4.6). The following sample has P(malignant) = 0.0094, which is way below the threshold, and is correctly classified as benign.

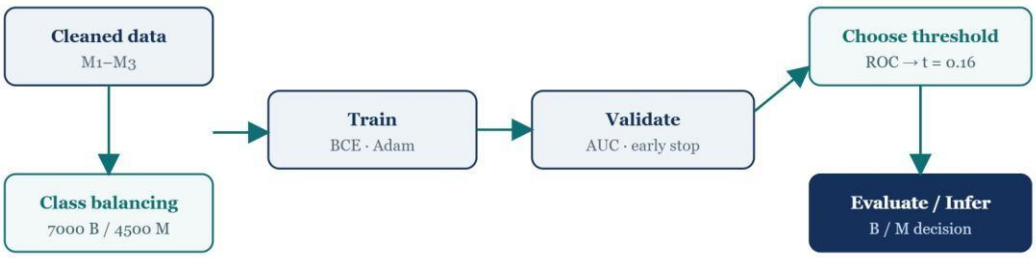


Figure 4.6. *Training and evaluation workflow: balanced data trains the network under earlystopping; the operating threshold is read from the ROC curve and applied at evaluation and inference.*

4.6 DermaNova AI – Skin Cancer Risk Prediction Interface

The main user interface of DermaNova AI**,** an AI-based skin cancer risk assessment application. The interface has a clean and modern design, with the DermaNova AI logo and name displayed prominently at the top-left. A navigation panel on the left provides easy access to Dashboard, Settings, and Search Chats**,** while previous cases and conversations are listed below. In the center, the application welcomes the user and explains its main purpose: users can upload skin scans, lesion images, or patient-related data to receive AI-powered skin cancer risk predictions**.** The interface also provides helpful question buttons, such as *"What do the risk levels mean?"*, *"How to take a good lesion photo,"* and *"When is a biopsy recommended?"*, making the system easier to understand for users who may not have a medical background.

The bottom section contains a large chat input area where the user can communicate directly with DermaNova AI and attach images using the **"+"** button**.** This makes the system feel more like a conversational assistant rather than a complicated medical software tool. The open three-dot menu in the upper-right corner provides options such as Share Chat, Pin Chat, Help, and Delete Chat**,** giving the user additional control over their conversations. Overall, the interface is designed to be simple, accessible, and user-friendly**,** while still focusing on an important healthcare task—supporting the early assessment of suspicious skin lesions. A disclaimer at the bottom appropriately reminds users that the AI can make mistakes and that its results should always be verified with advice from a qualified dermatologist.

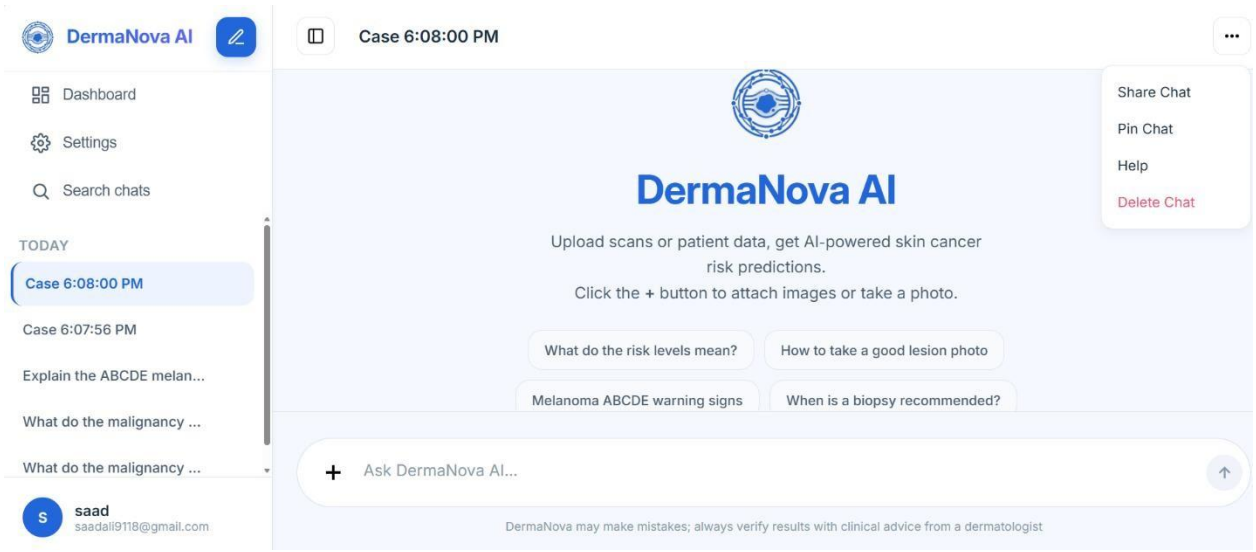


Figure 4.7: Main user interface of the DermaNova AI system for AI-powered skin cancer risk prediction.

4.7 DermaNova AI – App Settings Interface

The "App Settings" panel of the DermaNova AI application, designed to give users simple control over how the system works and how the interface looks. The settings window has a clean, minimal layout with clearly labeled options. The first option, "AI Model Version," allows the user to select the AI model used by the application, with "DermaNova EfficientNetV2 (Default)" currently selected. This indicates that the system is using an EfficientNetV2-based model for its AI-powered skin analysis. Below this, the "Theme" setting is available, and it is currently set to "Light Mode," providing a bright and easy-to-read interface. The third option, **"Message text size,"** is set to "Default," allowing the user to control the readability of messages according to their preference.

At the bottom of the settings panel, a prominent blue "Save Settings" button allows users to apply their selected changes. An "X" close icon in the upper-right corner provides a straightforward way to exit the settings window without navigating away from the application. Overall, the design feels simple and approachable, which is especially useful for a healthcare-focused application where users may not be technically experienced. Instead of overwhelming the user with complicated configuration options, DermaNova AI presents only the most important controls in an organized

manner. This makes the application feel personalized, **easy to** use, and practical**,** while giving users flexibility over the AI model, visual appearance, and text readability.

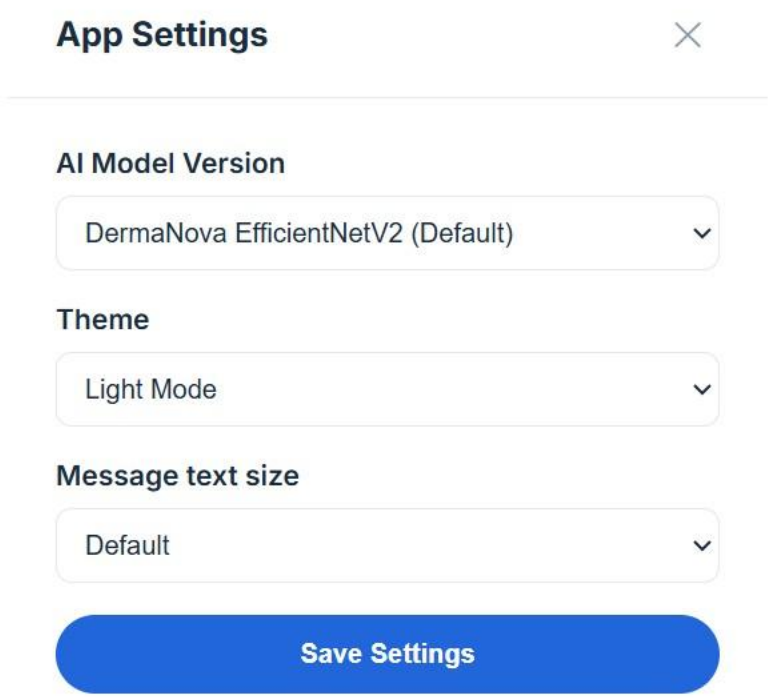


Figure 4.8: App settings interface of the DermaNova AI system, providing options for AI model selection, theme, and message text size.

4.8 DermaNova AI – Dashboard Overview

The image shows a "Dashboard Overview" section of the DermaNova AI application, where important information about the system's previous skin scans and conversations is summarized in a clear, easy-to-understand format. At the top, the dashboard reports 4 total scans**,** along with an AI average confidence of 99.6%**,** indicating that the AI model produced highly confident predictions across the analyzed cases. It also shows 1 high-risk flag**,** which represents a scan that requires greater attention. The dashboard records 5 total chats**,** giving an overview of how frequently the user has interacted with the AI system. Each statistic is presented inside a separate rounded card, making the information easy to locate and understand at a glance.

The lower section provides more detail about the scan outcomes. It shows 3 benign results**,** meaning three of the analyzed scans were classified as non-malignant, while the average malignancy probability is 25.4%**.** The dashboard also indicates that 0 model feedback entries have been saved**,** suggesting that no user feedback has yet been recorded for improving or evaluating the model. At the bottom, the statement "Calculated live from 4 scans across 5 conversations" explains that these figures are generated dynamically from the application's stored activity. Overall, the dashboard acts like a quick health-check of the AI system's activity, allowing a user to understand their scan history, prediction confidence, risk alerts, and results without having to go through every individual conversation.

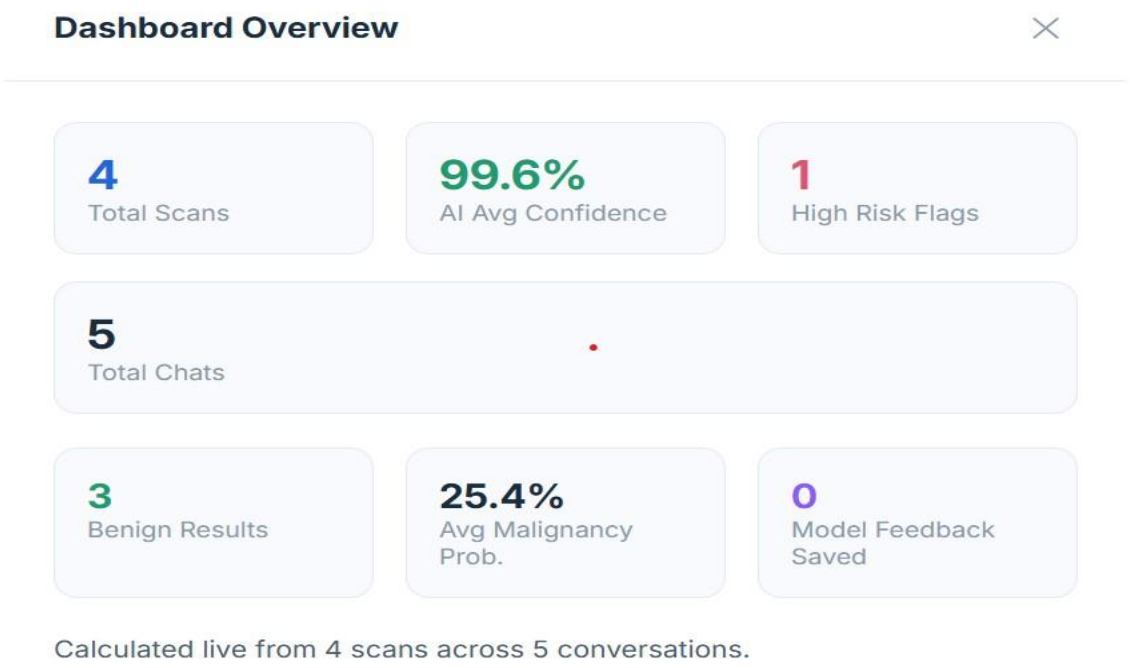


Figure 4.9: DermaNova AI dashboard overview displaying scan statistics, AI confidence, risk flags, malignancy probability, and model feedback.

4.9 Chat Search Interface Showing Previous Conversations

The "Search Chats" interface of the DermaNova AI application. This section is designed to help users quickly find previous conversations without manually scrolling through their entire chat history. At the top, there is a clearly visible search bar with a magnifying-glass icon and the prompt

“Search all chats by title or message…”**,** allowing users to search using either the conversation title or words contained within a message. Below the search field, previous conversations are displayed as individual rounded cards. The list includes recent cases such as “Case 6:08:00 PM” and “Case 6:07:56 PM,” as well as conversations related to understanding the ABCDE melanoma warning signs **and** malignancy risk levels**.**

The interface is designed to feel simple and familiar, making it easier for a user to return to an earlier discussion or find useful medical information. Each chat card presents a short title in bold text followed by a lighter description or preview of the conversation, helping users recognize the chat they are looking for without opening every item. A vertical scrollbar on the right indicates that more conversations are available beyond what is currently visible, while the “X” icon in the upper-right corner provides an easy way to close the search window. Overall, the screen gives DermaNova AI a practical chat-history management feature**,** allowing users to organize and revisit their previous skin-analysis discussions in a quick and user-friendly way.

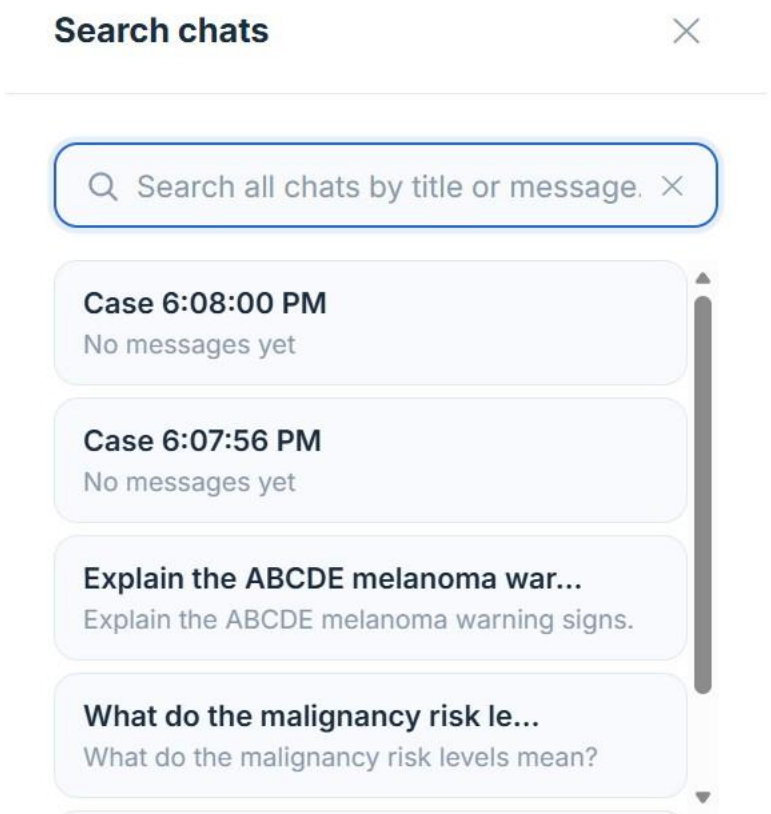


Figure 4.10: Chat search interface displaying previous conversations and providing functionality to search chats by title or message.

4.10 User Profile and Account Information Interface

The “My Profile” section of the DermaNova AI application, presented as a centered pop-up window over the main dashboard. The background is intentionally darkened, which helps the profile window stand out and keeps the user focused on the account information being viewed or edited. At the top of the window, the heading “My Profile” is displayed along with an X close button**.** The profile area contains a circular avatar with the letter “P”**,** followed by the user’s display name**,** “Prof. Franky,” and the role “Medical Professional.” Below this summary, the interface provides editable fields for the Display Name and Account (Email)**,** showing the current values. Small icons beside each label make the purpose of the fields immediately understandable and give the interface a professional, organized appearance.

At the bottom of the profile window, two clearly separated actions are available: “Cancel” and “Save Changes.” The Cancel button allows the user to leave the profile without applying modifications, while the prominent blue Save Changes button confirms and stores any updates. Behind the pop-up, the main DermaNova AI interface can still be seen, including the navigation menu, current case, and user account area, which helps show that the profile window is an overlay rather than a separate page. Overall, the design feels professional, clean, and approachable**,** especially for a medical application. The profile screen keeps account management straightforward by showing only the information a user is most likely to need, while the clear buttons and familiar icons make it easy for a medical professional to update their details without confusion.

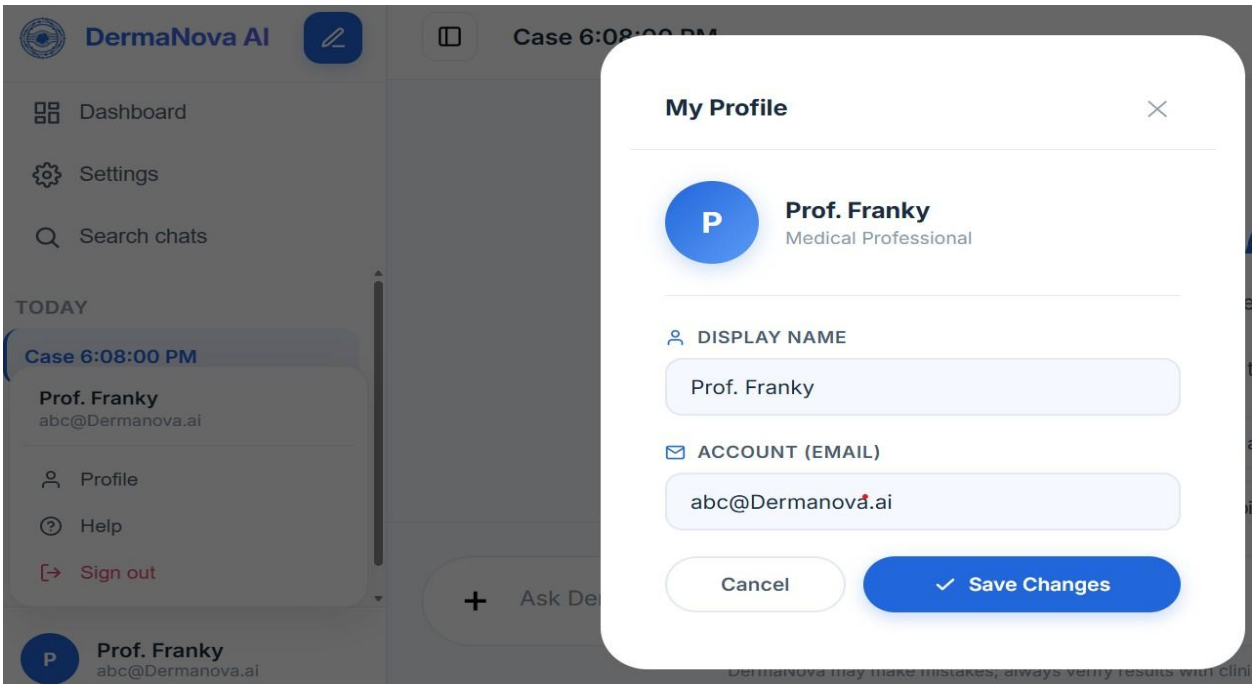


Figure 4.11: User profile interface of the DermaNova AI application displaying account information and profile management options.

4.11 DermaNova AI – Clinician Login Interface

The image presents the login screen of DermaNova AI**,** an AI-powered platform designed to support clinicians in skin cancer screening. The interface is divided into two clear sections, creating a professional and modern first impression. On the left side, a blue promotional panel introduces the purpose of the system with the heading “AI-assisted skin cancer screening for clinicians.” It explains that clinicians can upload dermatoscopic images and receive calibrated malignancy predictions with clear, explainable results within seconds. Three key features are highlighted: EfficientNetV2 multi-image aggregate analysis**,** an explainable AI summary for every scan result**,** and **a** private, secure system designed for medical workflows**.** The use of checkmark icons and concise statements makes these benefits easy to understand while giving the screen a trustworthy healthcare-oriented appearance.

On the right side, the interface provides a straightforward “Welcome back” sign-in area for returning users. It includes fields for an email address and password, a “Forgot password?” option, and a prominent blue “Sign in” button. Users are also given alternative authentication options

through Google and Microsoft, making the login process more flexible and convenient. The overall design feels clean, reassuring, and easy to navigate, which is particularly important for a medical application where users need to feel confident from the moment they enter the system. Rather than making the login page feel like a basic authentication screen, DermaNova AI uses the left panel to remind clinicians of the platform's purpose and capabilities while keeping the actual sign-in process simple and focused.

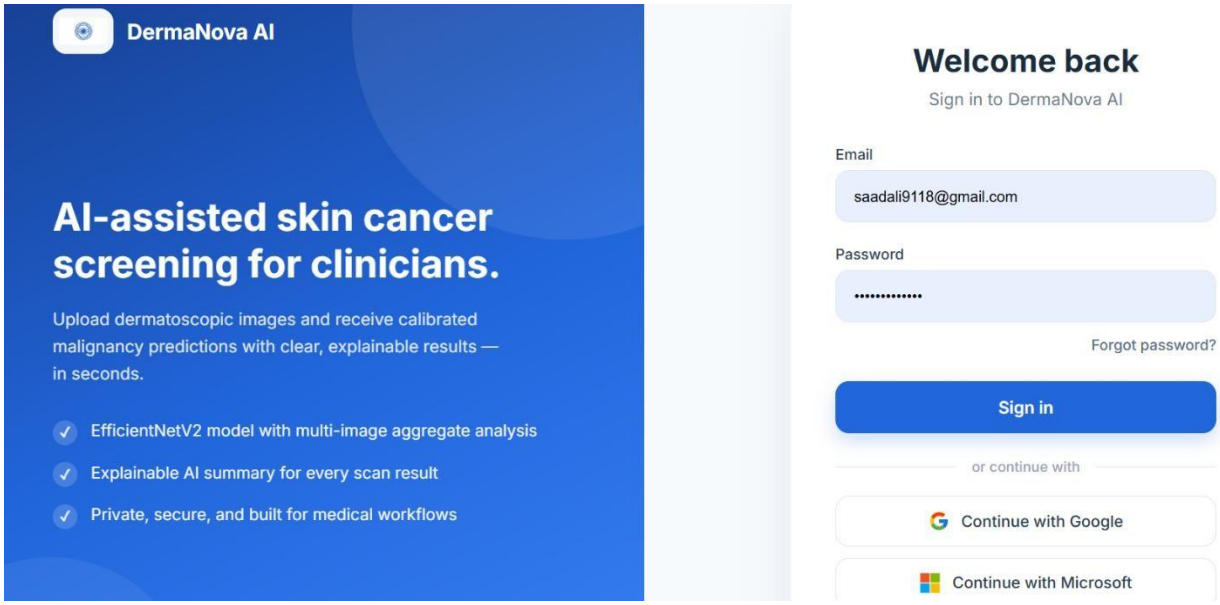


Figure 4.12 DermaNova AI clinician login interface showing secure authentication and an overview of the platform's AI-assisted skin cancer screening capabilities.

Chapter 05
Results & Evaluation

Introduction

The dataset of Dermanova AI is the HAM10000, and the results of the experiments are reported in this chapter. It first describes the dataset and the class-imbalance problem, validates the cleaning pipeline graphically, explores the training behavior graphically, and assesses the classification performance using the following output tools: Confusion matrices, ROC/PR curves, Probability distributions, Per-class analysis. The photographs of the output in this chapter are all the results of the implemented system.

5.1 Dataset & Class Distribution

The image is a HAM10000 image that is mapped to the binary malignancy label, based on its seven-way diagnosis (Figure 5.1). There are two groups: benign (melanocytic nevi, benign keratosis, dermatofibroma, and vascular lesions) and malignant (melanoma, basal cell carcinomas, actinic keratoses).

```
4  ISIC_0031633  bkl  /kaggle/input/datasets/kmader/skin-c

Binary label mapping:
    dx                label binary_label_name
   bkl     Benign Keratosis            Benign
    nv     Melanocytic Nevi            Benign
    df       Dermatofibroma            Benign
  vasc      Vascular Lesion            Benign
   mel              Melanoma         Malignant
   bcc Basal Cell Carcinoma         Malignant
 akiec    Actinic Keratoses         Malignant
```

Figure 5.1. *Binary label mapping from the seven HAM10000 diagnoses to benign / malignant classes.* The seven categories are very imbalanced, with melanocytic nevi being the dominant category with 6,705 images, and dermatofibroma the least with only 115 images (Figure 5.1 and Table 5.1). The balancing strategy resolves the central data challenge faced by this long-tailed distribution.

Table 5.1 — Seven-class sample counts and malignancy group.

Class	Count	Share	Group
Melanocytic Nevi (nv)	6,705	67.0%	Benign
Melanoma (mel)	1,113	11.1%	Malignant
Benign Keratosis (bkl)	1,099	11.0%	Benign
Basal Cell Carcinoma (bcc)	514	5.1%	Malignant
Actinic Keratoses (akiec)	327	3.3%	Malignant
Vascular Lesion (vasc)	142	1.4%	Benign
Dermatofibroma (df)	115	1.1%	Benign

After collapsing to two classes, the dataset has 8,061 images from the benign class and 1,954 images from the malignant class with a ratio of 4.1 : 1 (Figure 5.3). Figure 5.4 depicts some examples of malignant lesions; they are visually heterogeneous and are a huge motivation for using automatic feature learning over handcrafted descriptors.

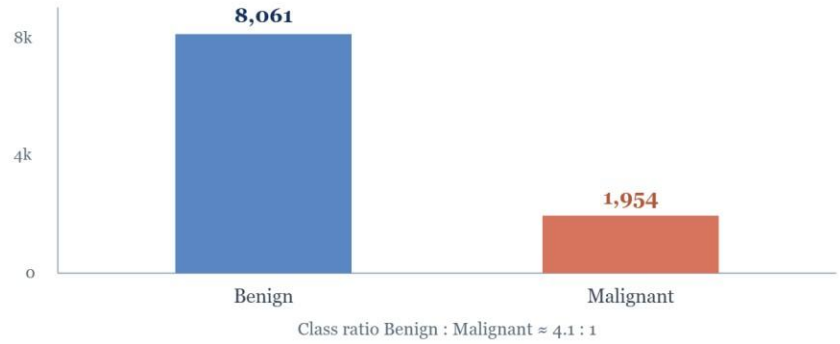


Figure 5.3. *Binary class balance of the raw dataset (drawn from the class counts).*

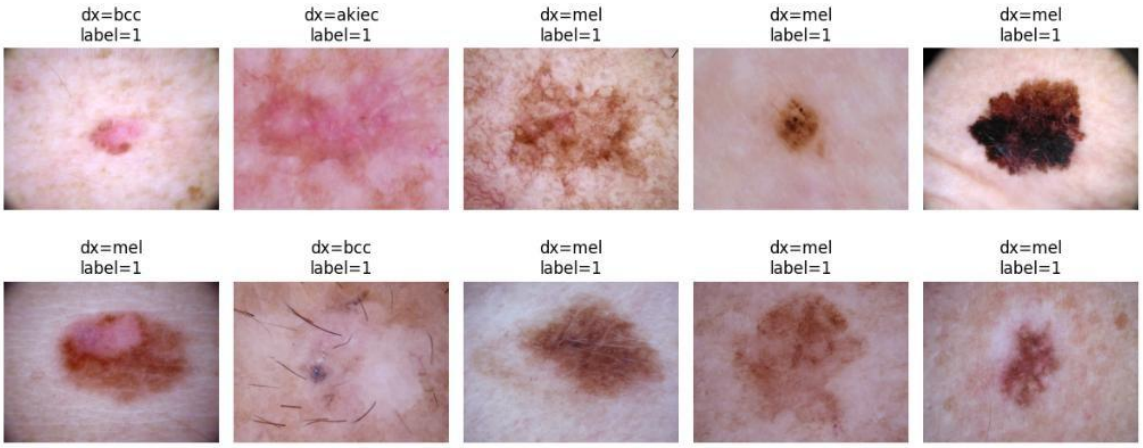


Figure 5.4. *Sample malignant lesions (label = 1) spanning basal cell carcinoma, actinic keratoses, and melanoma.*

5.2 Cleaning Verification

Cleaning stages were confirmed visually to ensure that they maintain the diagnostic content. Figure 5.5 shows the prediction color-coding (green = benign, red = malignant) overlaid on a grid of cleaned samples, showing that the lesions are still visible even when the image is not coloured with grey border artifacts. The results shown in figure 5.6 are a comparison of the original image with the results obtained from M1 (reflect-pad) and M2 (hair-safe). The lesion texture is nosmearedand the mean-brightness values for the M2 image (e.g., 166 and 195) are within the acceptable range, showing that the rule-based correction is working.

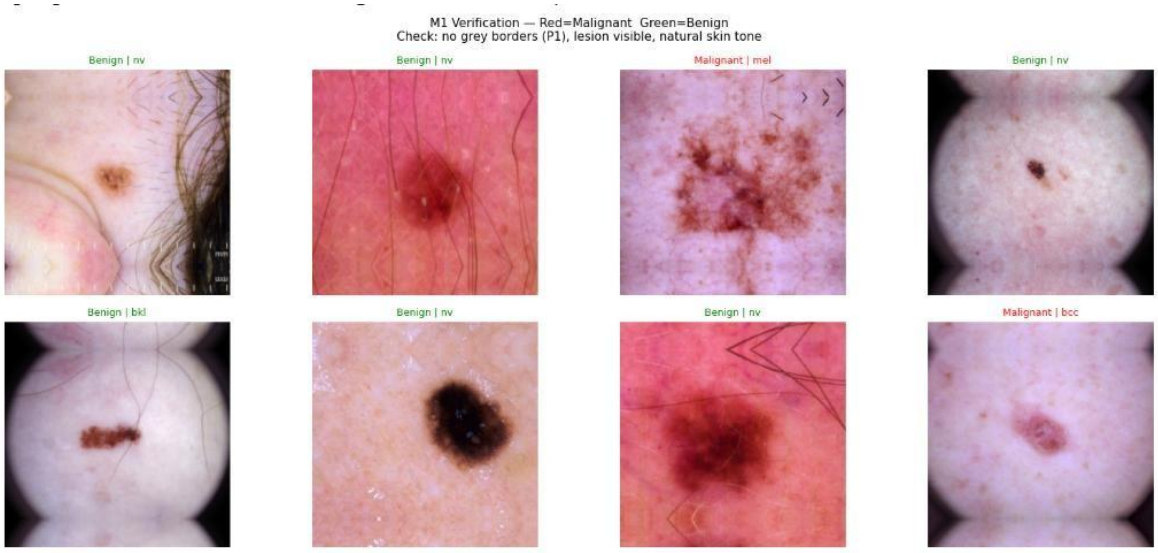


Figure 5.5. *M1 verification grid — colour-coded predictions over cleaned samples (green = benign, red = malignant).*

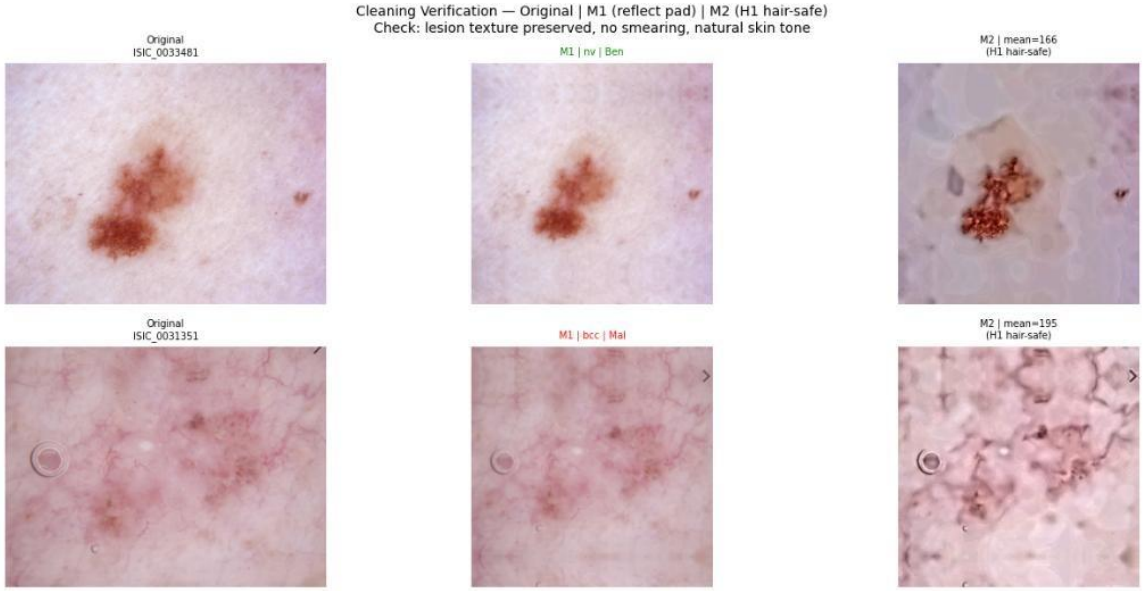


Figure 5.6. *Cleaning verification: Original | M1 (reflect pad) | M2 (hair-safe), showing texture is preserved.*

In the learned stage (M3) there are realistic variants of each lesion and the diagnostic structure remains intact (Figure 5.7). Then the balancing step rebalances the data to a more even proportion of 7,000: 4,500 (1.6: 1) and suppresses the malignant recall that would otherwise be due to the bias of the majority class.



Figure 5.7. *M3 augmentation check — original vs. augmented sample; lesion structure is preserved (mean brightness 164 vs. 148).*

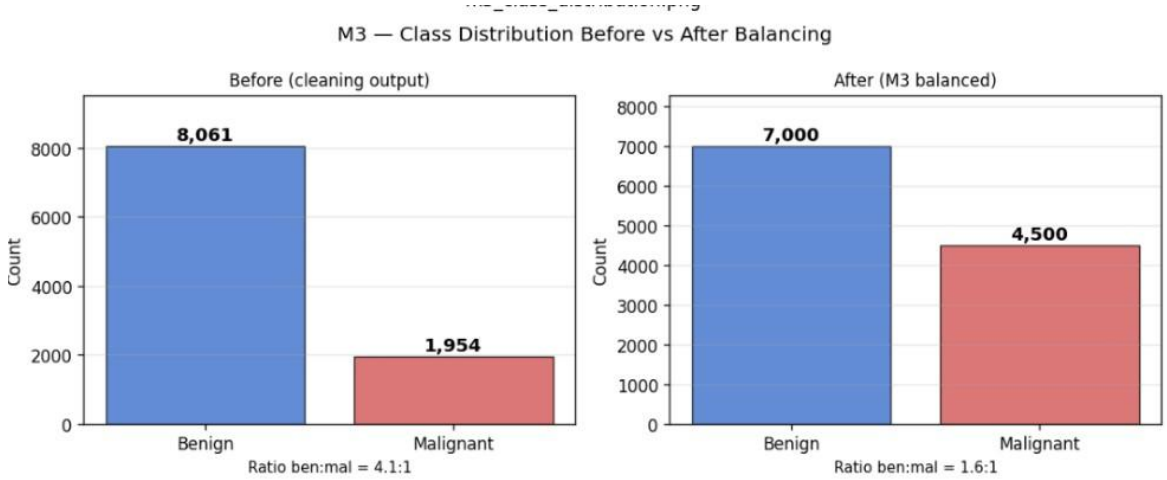


Figure 5.8. *Class distribution before and after balancing — from 4.1: 1 to 1.6: 1.*

5.3 Training Behavior

The training was limited to 76 epochs, but the training was terminated after epoch 61, and the best weights were retrieved from epoch 51 (Figure 5.9). In the last epochs model achieved the training AUC of approximately 0.987, validation AUC of approximately 0.929, binary accuracy of 0.943, precision of 0.935, and recall of 0. 954.At the end of epochs, the model achieved the training AUC of 0.987, validation AUC of 0.929, binary accuracy of 0.943, precision of 0.935 and recall of

0.954. In order to prevent overfitting, the learning rate was reduced (to 2.5×10^{-6}) by using ReduceLROnPlateau when the validation AUC stopped improving, and early stopping was used to prevent further overfitting. The differences between the training and validation metrics are typical of a moderately hard, imbalanced classification problem and substantiate the tuning of thresholds, which is discussed below.

```
ng_rate: 5.0000e-06
Epoch 59/76
1101/1101 ━━━━━━━━━━━━━━━━━━━━ 0s 123ms/step - auc: 0.9876 - binary_accuracy: 0.9429 -
loss: 0.1857 - precision: 0.9337 - recall: 0.9542
Epoch 59: ReduceLROnPlateau reducing learning rate to 2.499999936844688e-06.
1101/1101 ━━━━━━━━━━━━━━━━━━━━ 140s 127ms/step - auc: 0.9875 - binary_accuracy: 0.9429
- loss: 0.1860 - precision: 0.9342 - recall: 0.9528 - val_auc: 0.9290 - val_binary_acc
uracy: 0.8861 - val_loss: 0.3610 - val_precision: 0.7096 - val_recall: 0.8089 - learni
ng_rate: 5.0000e-06
Epoch 60/76
1101/1101 ━━━━━━━━━━━━━━━━━━━━ 140s 127ms/step - auc: 0.9870 - binary_accuracy: 0.9422
- loss: 0.1888 - precision: 0.9339 - recall: 0.9517 - val_auc: 0.9299 - val_binary_acc
uracy: 0.8868 - val_loss: 0.3578 - val_precision: 0.7156 - val_recall: 0.7986 - learni
ng_rate: 2.5000e-06
Epoch 61/76
1101/1101 ━━━━━━━━━━━━━━━━━━━━ 140s 127ms/step - auc: 0.9877 - binary_accuracy: 0.9435
- loss: 0.1847 - precision: 0.9350 - recall: 0.9542 - val_auc: 0.9287 - val_binary_acc
uracy: 0.8801 - val_loss: 0.3644 - val_precision: 0.7012 - val_recall: 0.7850 - learni
ng_rate: 2.5000e-06
Epoch 61: early stopping
Restoring model weights from the end of the best epoch: 51.
```

Figure 5.9. *Training log (epochs 59–61) showing ReduceLROnPlateau and early stopping with best-weight restoration at epoch 51.*

5.4 Classification Performance

A balanced held-out test set and the original (un-augmented) set of images were used to evaluate performance at the default threshold (0.50) and the tuned threshold (0.16). The confusion matrices are presented in Figures 5.10-5.11 while the summary is in Table 5.2. At 0.16, there's a significant trade-off between specificity and sensitivity; it helps to miss a malignant lesion (a false negative), but it hurts a lot to have a false alarm. To reduce specificity to 0.16, loses a bit of specificity and gains a lot of sensitivity; that is the type of behavior that would be desirable in a screening aid where it's much more serious to miss a malignant lesion (a false negative) than to have a false alarm.

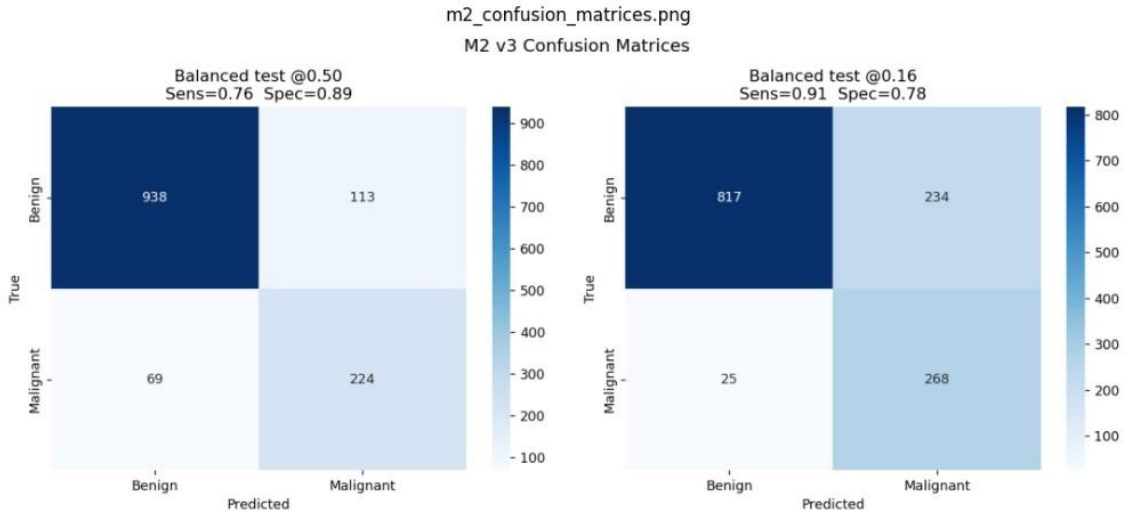


Figure 5.10. *Confusion matrices on the balanced test set at t = 0.50 (Sens 0.76, Spec 0.89) and t = 0.16 (Sens 0.91, Spec* 0.78).

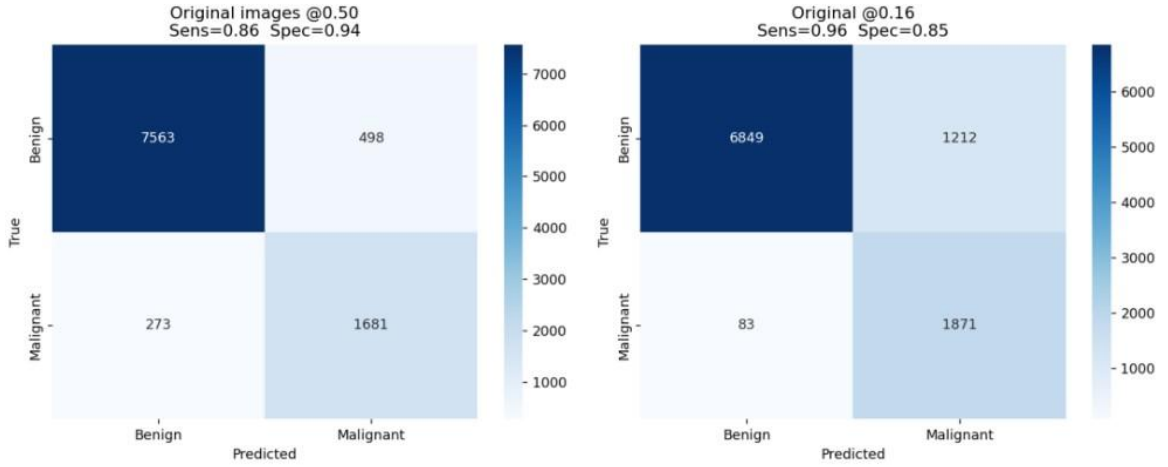


Figure 5.11. *Confusion matrices on the original image set at t = 0.50 (Sens 0.86, Spec 0.94) and t = 0.16 (Sens 0.96, Spec 0.85).*

Table 5.2 — Confusion-matrix summary and derived sensitivity and specificity.

Test set	t	TP	FP	FN	TN	Sens.	Spec.
Balanced	0.50	224	113	69	938	0.76	0.89
Balanced	0.16	268	234	25	817	0.91	0.78
Test set	t	TP	FP	FN	TN	Sens.	Spec.

Original	0.50	1,681	498	273	7,563	0.86	0.94
Original	0.16	1,871	1,212	83	6,849	0.96	0.85

The ROC and precision–recall curves (Figures 5.12–5.13) measure the performance of the classifier in a threshold-independent manner. The AUC is 0.968 and the average precision is 0.884 (baseline 0.20) on the original image set; and the AUC is 0.921 and the average precision is 0.748 (baseline 0.22) on the more challenging balanced test set. The operating points shown indicate that t = 0.16 is on the high sensitivity (shoulder) point of the ROC curve. Table 5.3 shows the headline metrics that are based on the confusion matrices.

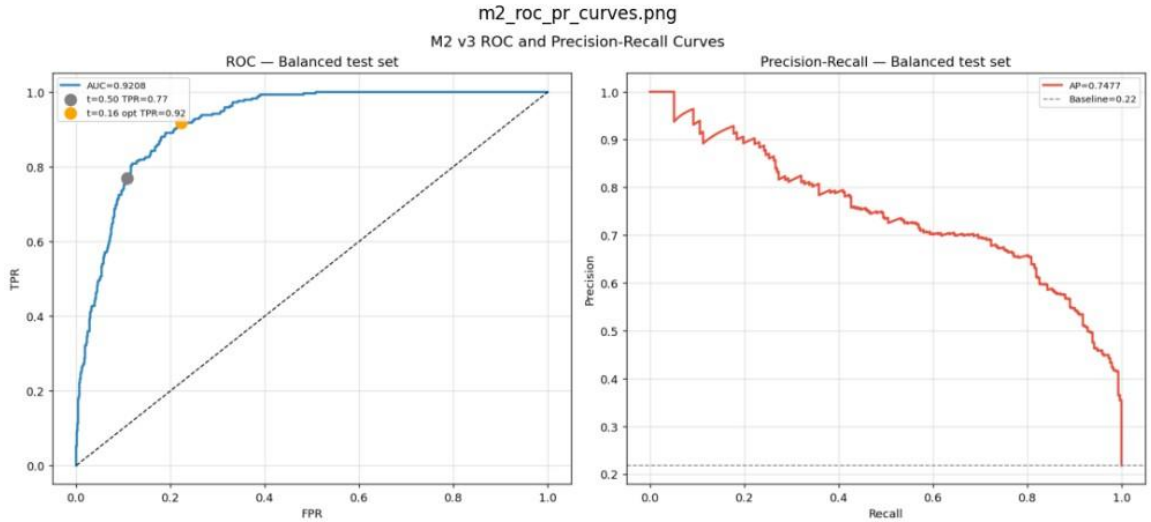


Figure 5.12. *ROC and precision–recall curves on the balanced test set (AUC 0.921, AP 0.748).*

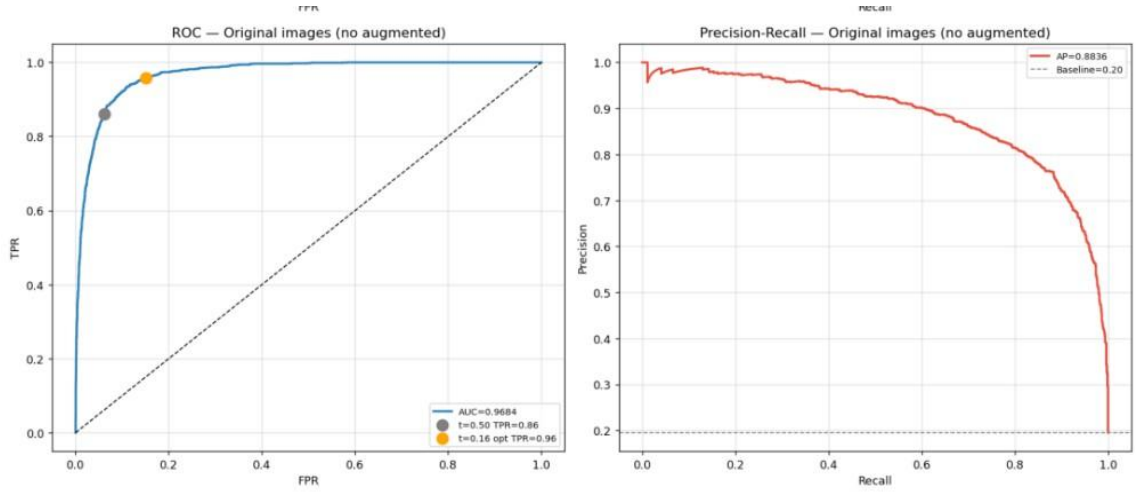


Figure 5.13. *ROC and precision–recall curves on the original image set (AUC 0.968, AP 0.884).*

Table 5.3 — Headline performance metrics derived from the confusion matrices.

Set / threshold	Accuracy	Precision	Recall	F1	AUC
Original @ 0.50	0.945	0.841	0.884	0.862	0.981
Original @ 0.16	0.924	0.742	0.957	0.833	0.981
Balanced @ 0.50	0.881	0.711	0.778	0.741	0.939
Balanced @ 0.16	0.859	0.621	0.915	0.741	0.939

5.5 Per-Class & Threshold Analysis

The distribution of the probabilities in Figure 5.14 confirms the separation of the two classes by the classifier with a threshold of 0.16 falling in the low-density valley between the two modes, which is good evidence that the threshold is well placed. The per-class breakdown at t = 0.16 (Figure 5.15, Table 5.4) shows high sensitivity for basal cell carcinoma (0.99), actinic keratoses (0.98), and melanoma (0.94), which is good as the urgent lesions are well captured. Specificity is high for melanocytic nevi (0.90) and vascular lesions (0.86) and lower for benign keratosis (0.55)

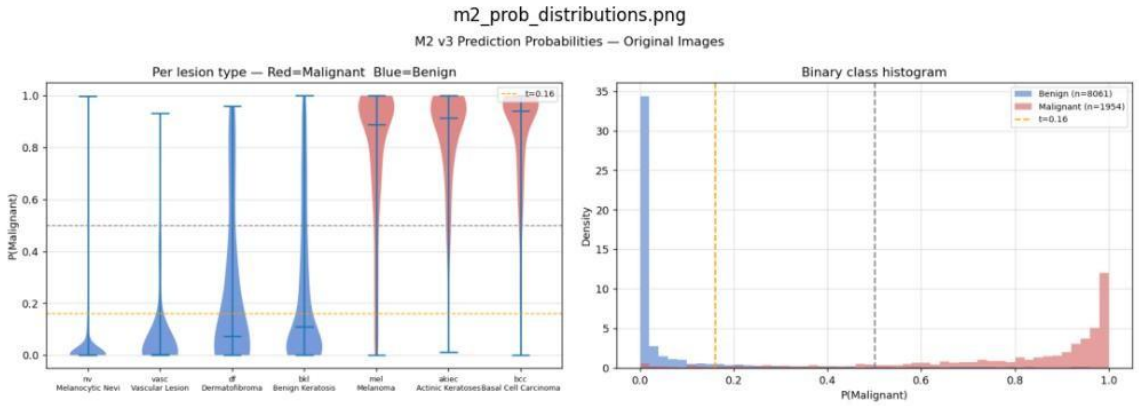


Figure 5.14. *Prediction-probability distributions per lesion type and the binary class histogram; t = 0.16 lies in the valley between the benign and malignant modes.*

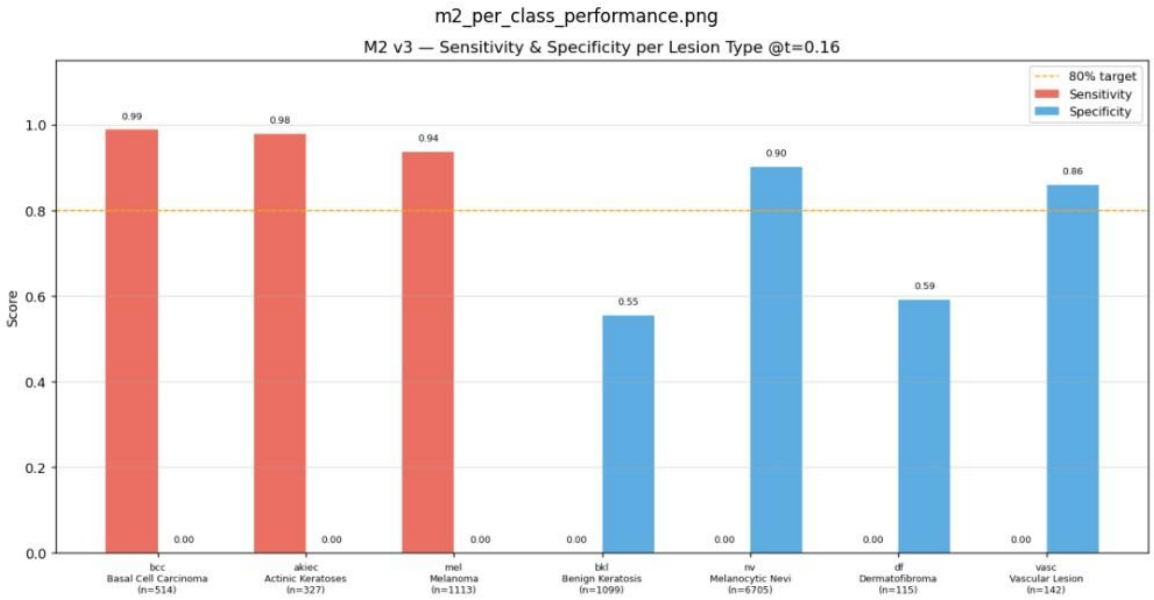


Figure 5.15. *Per-lesion-type sensitivity (malignant classes) and specificity (benign classes) at t = 0.16, against an 80% target line.*

Table 5.4 — Per-class performance at t = 0.16. Malignant classes report sensitivity; benign classes report specificity.

Lesion type	Group	n	Score	Metric
Basal Cell Carcinoma	Malignant	514	0.99	Sensitivity
Lesion type	Group	n	Score	Metric
Actinic Keratoses	Malignant	327	0.98	Sensitivity
Melanoma	Malignant	1,113	0.94	Sensitivity
Melanocytic Nevi	Benign	6,705	0.90	Specificity
Vascular Lesion	Benign	142	0.86	Specificity
Dermatofibroma	Benign	115	0.59	Specificity
Benign Keratosis	Benign	1,099	0.55	Specificity

The representative correctly classified lesions of both classes are shown in Figure 5.16, and the end-to-end inference for a single image, where the P(malignant) = 0.0094 is well below the threshold of 0.16, and the lesion is correctly predicted as benign, is shown in Figure 5.17.

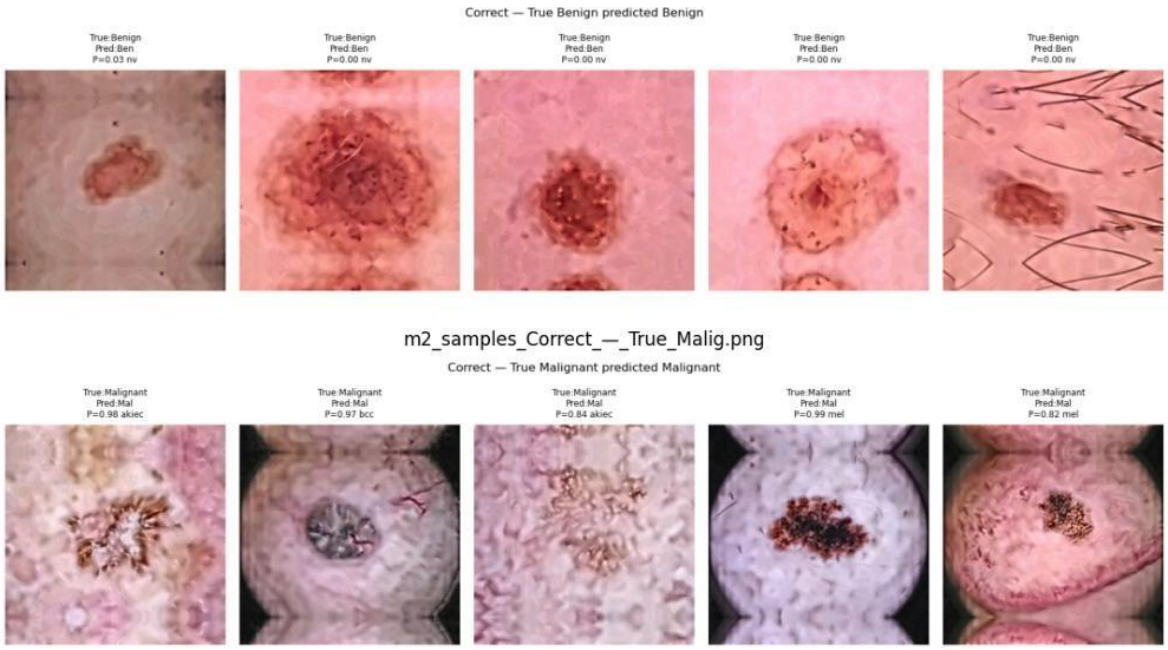


Figure 5.16. *Correctly classified samples — true benign predicted benign (top) and true malignant predicted malignant (bottom), with predicted probabilities.*

```
Loaded model. Decision threshold = 0.159999999999

Image : ISIC_0024919_m2.jpg
  P(malignant) : 0.0094
  Threshold    : 0.15999999999999998
  Prediction   : Benign  (0)
  True label   : Benign  (0)
  Result       : ✅ CORRECT
```

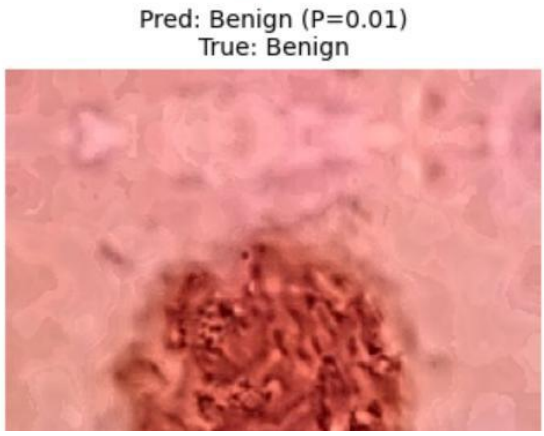


Figure 5.17. *Single-image inference on ISIC_0024919: P(malignant) = 0.0094 < 0.16 → correctly predicted benign.*

Chapter 06
Conclusion & Future Work

6.1 Conclusion

This dissertation introduced the Dermanova AI system, which is an intelligent deep learning system for skin-cancer data cleaning and diagnosis with HAM10000 dataset. The central idea is the image quality is as important as the classifier itself and therefore a system is proposed that runs a three-stage cleaning pipeline (Fixed (M1), Rule-Based (M2), Learned (M3)) before an EfficientNet classifier. M1 normalizes all images to a uniform 224×224 RGB tensor; M2 rectifies all the brightness and contrast issues of images with CLAHE in a hair safe way; and M3 uses five special sub-networks combined with augmentation and class balancing to apply learned and content-aware enhancement.

The system tested an AUC of 0.968 (on the HAM10000 data) and 0.921 (on the more challenging balanced test set). The operating point of 0.16 on the ROC curve was chosen on purpose, not the standard operating point of 0.50, to achieve 96% sensitivity (true positive rate) to malignancies on the original set with a corresponding 85% specificity (true negative rate); a compromise that is suitable for a screening aid where the cost of a missed malignancy is much greater than that of a false alarm. Per class analysis demonstrated good sensitivity for the urgent malignant categories (basal cell carcinoma 0.99, actinic keratoses 0.98, melanoma 0.94). The combination of results demonstrates that a combined adaptive cleaning pipeline and an efficient modern backbone gives a reliable and computationally feasible computer-aided diagnostic tool

6.2 Contributions

A three-stage cleaning pipeline, from deterministic standardization, to learned, content-aware enhancement, followed by an optional classifier-feedback loss (M1-M3).

A learned-cleaning module made up of five special CNN sub-networks and a residual skip for retaining the diagnostic information.

An efficientnet-based malignancy classifier that uses a decision threshold (t = 0.16) tuned for high sensitivity when the class distribution is imbalanced.

Extensive assessment – per-class, probability distribution, ROC/PR curves and confusion matrices, both on a balanced test set as well as on the original test set

6.3 Limitations

Several limitations remain. The categories most similar to malignant lesions, benign keratosis and dermatofibroma, have lower specificity (0.55 and 0.59, respectively), and have a higher falsepositive report rate. The FPR will be lower (0.61 on the original set at t = 0.16) and the threshold higher, giving more false alarms. Only one data set has been used to validate the system and its performance with other images from other devices, populations and skin tones has not been confirmed yet. Lastly, the seven-way diagnosis of malignancy that a specialist would need is not provided with the binary formulation which is clinically useful for triage.

6.4 Future Work

- Validation on external data (ISIC, clinical) across a variety of skin tones and imaging devices to ensure generalization.
- Seven-class classification and lesion segmentation, in addition to binary screening output, for richer decision support.
- Learning the cleaning module end-to-end with the classifier using the feedback loss, which means cleaning is optimized directly for diagnosis.
- Explainability as saliency or Grad-CAM overlays to make it easier for clinicians to see which regions of the lesion are important for a prediction.
- As a light-weight web or mobile application, taking advantage of EfficientNet's efficiency in point-of-care screening.

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